

Immunology

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3.1

Basic Principles of Immunology

1. **From the following, identify a specific component of the adaptive immune system that is formed in response to antigenic stimulation:**

A. Lysozyme
B. Complement
C. Commensal organisms
D. Immunoglobulin (Ig)

Immunology/Apply knowledge of fundamental biological characteristics/Immune system/1

2. **Which two organs are considered the primary lymphoid organs in which immunocompetent cells originate and mature?**

A. Thyroid and Peyer patches
B. Thymus and bone marrow
C. Spleen and mucosal-associated lymphoid tissue (MALT)
D. Lymph nodes and thoracic duct

Immunology/Apply knowledge of fundamental biological characteristics/Immune system/Organs/1

3. **What type of B cells is formed after antigen stimulation?**

A. Plasma cells and memory B cells
B. Mature B cells
C. Antigen-dependent B cells
D. Receptor-activated B cells

Immunology/Apply knowledge of fundamental biological characteristics/Immune system/Cells/1

4. **T cells travel from bone marrow to the thymus for maturation. What is the correct order of the maturation sequence for T cells in the thymus?**

A. Bone marrow to the cortex; after thymic education, released back to peripheral circulation
B. Maturation and selection occur in the cortex; migration to the medulla; release of mature T cells to secondary lymphoid organs
C. Storage in either the cortex or medulla; release of T cells into the peripheral circulation
D. Activation and selection occur in the medulla; mature T cells are stored in the cortex until activated by antigen

Immunology/Apply knowledge of fundamental biological characteristics/Immune system/Cells/1

Answers to Questions 1–4

1. **D** Ig is a specific part of the adaptive immune system and is formed only in response to a specific antigenic stimulation. Complement, lysozyme, and commensal organisms all act nonspecifically as a part of the adaptive immune system. These three components

do not require any type of specific antigenic stimulation.

2. **B** Bone marrow and the thymus are considered primary lymphoid organs because immunocompetent cells either originate from them or mature in them. Some immunocompetent cells mature or reside in bone marrow (the source of all hematopoietic cells) until transported to the thymus, spleen, or Peyer patches, where they process antigen or manufacture antibody. T lymphocytes, after originating in bone marrow, travel to the thymus to mature and differentiate.
3. **A** Mature B cells exhibit surface Ig that may cross-link a foreign antigen, thus forming the activated B cell and leading to capping and internalization of antigen. The activated B cell gives rise to plasma cells that produce and secrete Igs and memory cells that reside in lymphoid organs.
4. **B** Immature T cells travel from bone marrow to the thymus to mature into functional T cells. Once in the thymus, T cells undergo a selection and maturation sequence that begins in the cortex and moves to the medulla of the thymus. Thymic factors, such as thymosin and thymopoietin, and cells within the thymus, such as macrophages and dendritic cells, assist in this sequence. After completion of the maturation cycle, T cells are released to secondary lymphoid organs to await antigen recognition and activation.

5. Which cluster of differentiation (CD) marker is the most specific identifying marker for mature T cells?

- A. CD1
- B. CD2
- C. CD3
- D. CD4 or CD8

Immunology/Apply principles of basic laboratory procedures/T cells/Markers/1

6. Which markers are found on mature, peripheral helper T cells?

- A. CD1, CD2, CD4
- B. CD2, CD3, CD8
- C. CD1, CD3, CD4
- D. CD2, CD3, CD4

Immunology/Apply knowledge of fundamental biological characteristics/T cells/Markers/1

7. Which T cells express the CD8 marker and act specifically to kill tumors or virally infected cells?

- A. Helper T cells
- B. Suppressor T cells
- C. Cytotoxic T cells (T_C cells)
- D. Regulator T cells

Immunology/Apply knowledge of fundamental biological characteristics/T cells/Cytokines/1

8. How are T_C cells and natural killer (NK) cells similar?

- A. Require antibody to be present
- B. Effective against virally infected cells
- C. Recognize antigen in association with human leukocyte antigen (HLA) class II markers
- D. Do not bind to infected cells

Immunology/Apply knowledge of fundamental biological

characteristics/Lymphocytes/Functions/1

9. What is the name of the process by which phagocytic cells are attracted to a substance, such as a bacterial peptide?

- A. Diapedesis
- B. Degranulation
- C. Chemotaxis
- D. Phagotaxis

Immunology/Apply knowledge of fundamental biological characteristics/Immune system/Cells/1

10. All of the following are immunologic functions of complement except:

- A. Induction of an antiviral state
- B. Opsonization
- C. Chemotaxis
- D. Anaphylatoxin formation

Immunology/Apply knowledge of fundamental biological characteristics/Complement/Functions/1

11. Which complement component is found in both the classic and alternative pathways?

- A. C1
- B. C4
- C. Factor D
- D. C3

Immunology/Apply knowledge of fundamental biological characteristics/Complement/Components/1

Answers to Questions 5–11

- 5. **C** The CD3 marker appears during the early stages of T-cell development and can be used to differentiate T cells from other lymphocytes.
- 6. **D** Mature, peripheral helper T cells have the CD2, CD3 (mature T cell), and CD4 (helper) markers.
- 7. **C** T_C cells recognize antigen in association with major histocompatibility complex (MHC) class I complexes and act against target cells that express foreign antigens. These include viral antigens and the HLAs that are the target of graft rejection.
- 8. **B** Both T_C and NK cells are effective against virally infected cells, and neither requires antibody to be present to bind to infected cells. NK cells do not exhibit MHC class restriction, whereas activation of T_C cells requires the presence of MHC class I molecules in association with the viral antigen.
- 9. **C** Chemotaxis is the process by which phagocytic cells are attracted toward an area where they detect a disturbance in the normal functions of body tissues. Products from bacteria and viruses, complement components, coagulation proteins, and cytokines from other immune cells may all act as chemotactic factors.
- 10. **A** Complement components are serum proteins that function in opsonization, chemotaxis, and anaphylatoxin formation but do not induce an antiviral state in target

cells. This function is performed by interferons.

- 11. D** C3 is found in both the classic and alternative (alternate) pathways of the complement system. In the classic pathway, C3b forms a complex on the cell with C4b2a that enzymatically cleaves C5. In the alternative pathway, C3b binds to an activator on the cell surface. It forms a complex with factor B called *C3bBb*, which, like C4b2a3b, can split C5.

12. Which Ig(s) help(s) initiate the classic complement pathway?

- A. IgA and IgD
- B. IgM only
- C. IgG and IgM
- D. IgG only

Immunology/Apply knowledge of fundamental biological characteristics/Complement/Activation/1

13. How is complement activity destroyed in vitro?

- A. Heating serum at 56°C for 30 minutes
- B. Keeping serum at room temperature of 22°C for 1 hour
- C. Heating serum at 37°C for 45 minutes
- D. Freezing serum at 0°C for 24 hours

Immunology/Apply knowledge of fundamental biological characteristics/Complement/Activation/1

14. What is the purpose of C3a, C4a, and C5a, the split products of the complement cascade?

- A. To bind with specific membrane receptors of lymphocytes and cause release of cytotoxic substances
- B. To cause increased vascular permeability, contraction of smooth muscle, and release of histamine from basophils
- C. To bind with membrane receptors of macrophages to facilitate phagocytosis and the removal of debris and foreign substances
- D. To regulate and degrade membrane cofactor protein after activation by C3 convertase

Immunology/Apply knowledge of fundamental biological characteristics/Complement/Anaphylatoxins/1

15. Which region of the Ig molecule can bind antigen?

- A. Fragment antigen binding (Fab)
- B. Fragment crystallizable (Fc)
- C. **Constant light** (C_L)
- D. **Constant heavy** (C_H)

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Structures/1

16. Which region determines whether an Ig molecule can fix complement?

- A. **Variable heavy** (V_H)
- B. **Constant heavy** (C_H)
- C. **Variable light** (V_L)
- D. **Constant light** (C_L)

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Structures/1

17. Which Ig class(es) has (have) a J-chain?

- A. IgM
- B. IgE and IgD
- C. IgM and surface IgA (sIgA)
- D. IgG3 and IgA

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Structures/1

Answers to Questions 12–17

- 12. **C** Both IgG and IgM are the Igs that help to initiate the activation of the classic complement pathway. IgM is, however, a more potent complement activator.
- 13. **A** Complement activity in serum, in vitro, is destroyed by heating serum at 56°C for 30 minutes. In test procedures where complement may interfere with the test system, it may be necessary to destroy complement activity in the test sample by heat inactivation.
- 14. **B** C3a, C4a, and C5a are split products of the complement cascade that participate in various biological functions, such as vasodilation and smooth muscle contraction. These small peptides act as anaphylatoxins, for example, effector molecules that participate in the inflammatory response to assist in the destruction and clearance of foreign antigens.
- 15. **A** Fab is the region of the Ig molecule that can bind antigen. Two Fab fragments are formed from hydrolysis of the Ig molecule by papain. Each consists of a light chain and the V_H and C_{H1} regions of the heavy chain. The variable regions of the light and heavy chains interact, forming a specific antigen-combining site.
- 16. **B** The composition and structure of the constant region of the heavy chain determine whether that Ig will fix complement. Fc is formed by partial Ig digestion with papain and includes the C_{H2} and C_{H3} domains of both heavy chains. The complement component C1q molecule will bind to the C_{H2} region of an IgG or IgM molecule.
- 17. **C** Both IgM and sIgA have a J-chain joining individual molecules together; the J-chain in IgM joins five molecules and the J-chain in sIgA joins two molecules.

18. Which Ig appears first in the primary immune response?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Functions/1

19. Which immunoglobulin appears in highest titer in the secondary response?

- A. IgG

- B. IgM
- C. IgA
- D. IgE

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Functions/1

20. Which Ig can cross the placenta?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Functions/1

21. Which Ig cross-links mast cells to release histamine?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Functions/1

22. All of the following are functions of Igs except:

- A. Neutralizing toxic substances
- B. Facilitating phagocytosis through opsonization
- C. Interacting with T_C cells to lyse viruses
- D. Combining with complement to destroy cellular antigens

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Functions/1

23. Which of the following cell surface molecules is classified as an MHC class II antigen?

- A. HLA-A
- B. HLA-B
- C. HLA-C
- D. HLA-DR

Immunology/Apply knowledge of fundamental biological characteristics/MHC/HLA antigens/1

Answers to Questions 18–23

- 18. B** The first antibody to appear in the primary immune response to an antigen is IgM. The titer of antiviral IgM (e.g., IgM antibody to cytomegalovirus [anti-CMV]) is more specific for acute or active viral infection than IgG and may be measured to help differentiate active infection from prior infection.
- 19. A** A high titer of IgG characterizes the secondary immune response. Consequently, IgG antibodies comprise about 80% of the total Ig concentration in normal serum.
- 20. A** IgG is the only Ig class that can cross the placenta. All subclasses of IgG can cross the placenta, but IgG2 crosses more slowly. This process requires recognition of the Fc

region of the IgG by placental cells. These cells take up the IgG from maternal blood and secrete it into fetal blood, providing humoral immunity to the neonate for the first few months after delivery.

- 21. **D** IgE is the Ig that cross-links with basophils and mast cells. IgE causes the release of such immune response modifiers as histamine and mediates an allergic immune response.
- 22. **C** T_C cells lyse virally infected cells directly, without requirement for specific antibody. The T_C cell is activated by viral antigen that is associated with MHC class I molecules on the surface of the infected cell. The activated T_C cell secretes several toxins, such as tumor necrosis factor, which destroy the infected cell and virions.
- 23. **D** The MHC region is located on the short arm of chromosome 6 and codes for antigens expressed on the surface of leukocytes and tissues. The MHC region genes control immune recognition; their products include the antigens that determine transplant rejection. HLA-DR antigens are expressed on B cells. HLA-DR2, -DR3, -DR4, and -DR5 antigens show linkage with a wide range of autoimmune diseases.

24. Which MHC class of molecule is necessary for antigen recognition by CD4-positive T cells?

- A. Class I
- B. Class II
- C. Class III
- D. No MHC molecule is necessary for antigen recognition

Immunology/Apply knowledge of fundamental biological characteristics/MHC/Functions/1

25. Which of the following are products of HLA class III genes?

- A. T-cell immune receptors
- B. HLA-D antigens on immune cells
- C. Complement proteins C2, C4, and factor B
- D. Ig V_L regions

Immunology/Apply knowledge of fundamental biological characteristics/MHC/Functions/1

26. What molecule on the surface of most T cells recognizes antigen?

- A. IgT, a four-chain molecule that includes the tau heavy chain
- B. MHC protein, a two-chain molecule encoded by the HLA region
- C. CD3, consisting of six different chains
- D. T-cell receptor (TcR), consisting of two chains: α -chain and β -chain

Immunology/Apply knowledge of fundamental biological characteristics/Functions/1

27. TcR is similar to Ig molecules in that it:

- A. Remains bound to the cell surface and is never secreted
- B. Contains V and C regions on each of its chains
- C. Binds complement
- D. Can cross the placenta and provide protection to a fetus

Immunology/Apply knowledge of fundamental biological characteristics/TcR/Functions/2

28. Toll-like receptors (TLRs) are found on which cells?

- A. T cells
- B. Dendritic cells

- C. B cells
- D. Large granular lymphocytes

Immunology/Apply knowledge of fundamental biological characteristics/Innate immune system/Toll-like receptors/1

Answers to Questions 24–28

- 24. **B** Helper T lymphocytes (CD4-positive T cells) recognize antigens only in the context of a class II molecule. Because class II antigens are expressed on macrophages, monocytes, and B cells, the helper-T-cell response is mediated by interaction with processed antigen on the surface of these cells.
- 25. **C** Complement components C2 and C4 of the classic pathway and factor B of the alternative pathway are class III molecules. HLA-A, HLA-B, and HLA-C antigens are classified as class I antigens, and HLA-D, HLA-DR, HLA-DQ, and HLA-DP antigens are classified as class II antigens.
- 26. **D** T cells have a membrane bound receptor (TcR) that is antigen specific. This two-chain molecule consists of a single α -chain, similar to an Ig light chain, and a single β -chain, similar to an Ig heavy chain. Some T cells may express a γ - δ receptor instead of the α - β molecule. There is no τ heavy chain. MHC and CD3 molecules are present on T cells, but they are not the molecules that give antigen specificity to the cell.
- 27. **B** The antigen binding regions of both the α - and β -chains of the TcR are encoded by V genes that undergo rearrangement similar to that observed in Ig genes. The α -chain gene consists of V and J segments, similar to an Ig light chain. The β -chain consists of V, D, and J segments, similar to an Ig heavy chain. The α - and β -chains each have a single C-region gene encoding the constant region of the molecule. Although answer A is true for TcRs, it is not true for Igs that can be cell bound or secreted. Answers C and D are true for certain Ig heavy-chain isotypes but are not true for the TcR.
- 28. **B** TLRs are the primary antigen recognition protein of the innate immune system. They are found on antigen-presenting cells, such as dendritic cells and macrophages. Eleven TLRs have been described. TLRs recognize certain structural motifs common to infecting organisms. TLR 4, for example, recognizes bacterial lipopolysaccharide (LPS). The name TLR comes from its similarity to the Toll protein in *Drosophila*.

29. Macrophages produce which of the following proteins during antigen processing?

- A. IL-1 and IL-6
- B. γ -Interferon
- C. IL-4, IL-5, and IL-10
- D. Complement components C1 and C3

Immunology/Apply knowledge of fundamental biological characteristics/Innate immune system/Toll cytokines/2

30. A superantigen, such as toxic shock syndrome toxin-1 (TSST-1), bypasses the normal antigen-processing stage by binding to and cross-linking:

- A. A portion of an Ig molecule and complement component C1
- B. TLRs and an MHC class I molecule

- C. A portion of an Ig and a portion of a TcR
- D. A portion of a TcR and an MHC class II molecule

Immunology/Apply knowledge of fundamental biological characteristics/Antigen processing/Superantigens/2

31. T-regulator cells, responsible for controlling autoimmune antibody production, express which of the following phenotypes?

- A. CD3, CD4, CD8
- B. CD3, CD8, CD25
- C. CD3, CD4, CD25
- D. CD8, CD25, CD56

Immunology/Apply knowledge of fundamental biological characteristics/T cells/Markers/1

Answers to Questions 29–31

- 29. A** Interleukin-1 (IL-1) and IL-6 are proinflammatory macrophage-produced cytokines. In addition to their inflammatory properties, they activate T-helper cells during antigen presentation. γ -Interferon, IL-4, IL-5, and IL-10 are all produced by T cells. Complement components are produced by a variety of cells but are not part of the macrophage antigen-presentation process.
- 30. D** A superantigen binds to the V β -portion of the TcR and an MHC class II molecule. This binding can activate T cells without the involvement of an antigen-presenting cell. In some individuals, a single V β -protein that recognizes TSST-1 is expressed on up to 10% to 20% of T cells. The simultaneous activation of this amount of T cells causes a heavy cytokine release, resulting in the vascular collapse and pathology of toxic shock syndrome.
- 31. C** T-regulator cells are believed to be the primary immune suppressor cells and express CD3, CD4, and CD25. CD25 is the IL-2 receptor. CD25 may be expressed by activated T cells, but is constitutively expressed by the T-regulator cells. CD25 expression on T-regulator cells occurs in the thymus and is regulated by the FOXP3 protein.

3.2

Immunologic Procedures

- 1. The interaction between individual antigen and antibody molecules depends on several types of bonds, such as ionic bonds, hydrogen bonds, hydrophobic bonds, and van der Waals forces. How is the strength of this attraction characterized?**
 - A. Avidity
 - B. Affinity
 - C. Reactivity
 - D. Valency

Immunology/Apply principles of basic laboratory procedures/1

- 2. A laboratory is evaluating an enzyme-linked immunosorbent assay (ELISA) for detecting an antibody to cyclic citrullinated peptide (CCP), which is a marker for rheumatoid arthritis (RA). The laboratory includes serum from healthy volunteers and from patients with other connective tissue diseases in the evaluation. These specimens determine which factor of the assay?**
 - A. Sensitivity
 - B. Precision
 - C. Bias
 - D. Specificity

Immunology/Apply principles of basic laboratory procedures/RA/2

- 3. The detection of precipitation reactions depends on the presence of optimal proportions of antigen and antibody. A patient's sample contains a large amount of antibody, but the reaction in a test system containing antigen is negative. What has happened?**
 - A. Performance error
 - B. Low specificity
 - C. A shift in the zone of equivalence
 - D. Prozone phenomenon

Immunology/Apply principles of basic laboratory procedures/3

- 4. The positive and negative control values for an ELISA procedure are below their acceptable ranges. What is the most likely cause?**
 - A. Decay of the positive and negative controls
 - B. Incomplete washing following specimen addition
 - C. Overly long incubation times
 - D. Decay of the antibody–enzyme conjugate

Immunology/Apply principles of basic laboratory procedures/ELISA/3

Answers to Questions 1–4

- 1. B** Affinity refers to the strength of a single antibody–antigen interaction. Avidity is the strength of interactions between many different antibodies in a serum against a

particular antigen (i.e., the sum of many affinities).

2. **D** Specificity is defined as a negative result in the absence of the disease. The non-RA specimens would be expected to test negative if the assay has high specificity. Precision is the ability of the assay to repeatedly yield the same results on a single specimen. Both bias and sensitivity calculations would include specimens from RA persons. Although those specimens would be included in the evaluation, they are not listed in the question.
3. **D** Although performance error and low specificity should be considered, if a test system fails to yield the expected reaction, excessive antibody preventing a precipitation reaction is usually the cause. Prozone occurs when antibody molecules saturate the antigen sites, preventing cross-linking of the antigen-antibody complexes by other antibody molecules. Because antigen and antibody do not react at equivalence, a visible product is not formed, leading to a false-negative result.
4. **D** The antibody-enzyme conjugate is very sensitive to storage conditions and is easy to dissociate. Control specimens are unlikely to decay, and the other options would lead to higher values.

5. **What is the interpretation when an Ouchterlony plate shows crossed lines between wells 1 and 2 (antigen is placed in the center well and antisera in wells 1 and 2)?**

- A. No reaction between wells 1 and 2
- B. Partial identity between wells 1 and 2
- C. Nonidentity between wells 1 and 2
- D. Identity between wells 1 and 2

Immunology/Apply principles of basic laboratory procedures/Ouchterlony techniques/Interpretation/2

6. **A weight lifter taking many supplements is tested monthly for thyroid-stimulating hormone (TSH) in a direct capture assay, which uses a streptavidin-biotin indicator system. She has had normal TSH levels for the past 3 months on specimens collected in the late evening. This month she comes in right after breakfast for her blood draw. The TSH level is three times her previous level. What may be the cause of this difference?**

- A. Diurnal variation in TSH levels
- B. Exogenous biotin in her system from a supplement taken that morning
- C. Reduced thyroid function caused by an unidentified pathology
- D. Pipetting error

Immunology/Apply knowledge to identify sources of error/avidin-biotin assays/3

7. **What comprises the indicator system in an indirect ELISA for detecting antibody?**

- A. Enzyme-conjugated antibody + chromogenic substrate
- B. Enzyme conjugated antigen + chromogenic substrate
- C. Enzyme + antigen
- D. Substrate + antigen

Immunology/Apply principles of basic laboratory procedures/ELISA/1

8. **What outcome results from improper washing of a tube or well after adding the enzyme-antibody conjugate in an ELISA system?**

- A. Result will be falsely decreased
- B. Result will be falsely increased

- C. Result will be unaffected
- D. Result is impossible to determine

Immunology/Apply knowledge to identify sources of error/ELISA/3

9. What would happen if the color reaction phase is prolonged in one tube or well of an ELISA test?

- A. Result will be falsely decreased
- B. Result will be falsely increased
- C. Result will be unaffected
- D. Impossible to determine

Immunology/Apply knowledge to identify sources of error/ELISA/3

10. The absorbance of a sample measured by ELISA is greater than the highest standard. What corrective action should be taken?

- A. Extrapolate an estimated value from the highest reading
- B. Repeat the test using a standard of higher concentration
- C. Repeat the assay using one half the volume of the sample
- D. Dilute the test sample

Immunology/Evaluate laboratory data to take corrective action according to predetermined criteria/ELISA/3

Answers to Questions 5–10

- 5. **C** Crossed lines indicate nonidentity between wells 1 and 2. The antibody from well 1 recognizes a different antigenic determinant than the antibody from well 2.
- 6. **B** High levels of exogenous biotin may be present in blood up to 10 to 15 hours after ingestion and will usually cause falsely elevated values in avidin–biotin direct ELISA assays and competitive immunoassays but lower values in sandwich (immunometric) assays. Monthly variation in TSH should not approach the level seen in this situation.
- 7. **A** ELISA measures antibody by using immobilized reagent antigen. The antigen is fixed to the walls of a tube or bottom of a microtiter well. Serum is added (and incubated) and the antibody binds, if present. After washing, the antigen–antibody complexes are detected by adding an enzyme-labeled antiimmunoglobulin (anti-Ig). The unbound enzyme label is removed by washing, and the bound enzyme label is detected by adding chromogenic substrate. The enzyme catalyzes the conversion of substrate to a colored product.
- 8. **B** If unbound enzyme-conjugated anti-Ig is not washed away, it will catalyze the conversion of the substrate to a colored product, yielding a falsely elevated result.
- 9. **B** If the color reaction is not stopped within the time limits specified by the procedure, the enzyme will continue to act on the substrate, producing a falsely elevated test result.
- 10. **D** Usually, when a test sample reads at a value above the highest standard in an ELISA, the sample is diluted and measured again. In those instances where no additional clinical value can be obtained by dilution, the result may be reported as greater than the highest standard (citing the upper reportable limit of the assay).

11. A patient was suspected of having a lymphoproliferative disorder. After several laboratory tests were completed, the patient was found to have an IgM κ paraprotein. In what sequence should the laboratory tests leading to this diagnosis have been performed?

- A. Serum protein electrophoresis (SPE) followed by immunofixation electrophoresis (IFE)
- B. Ig levels, followed by SPE
- C. Total lymphocyte count, followed by Ig levels
- D. Ig levels, followed by urine protein electrophoresis

Immunology/Evaluate laboratory data to reach conclusions/Monoclonal gammopathy/3

12. An IFE performed on a serum sample showed a narrow dark band in the lanes containing anti- γ and anti- λ . How should this result be interpreted?

- A. Abnormally decreased IgG concentration
- B. Abnormal test result demonstrating monoclonal IgG λ
- C. Normal test result
- D. Impossible to determine without densitometric quantitation

Immunology/Evaluate laboratory data to make identifications/IFE/2

13. Which type of nephelometry is used to measure immune complex formation almost immediately after reagent has been added?

- A. Rate
- B. Endpoint
- C. Continuous
- D. One-dimensional

Immunology/Apply principles of basic laboratory procedures/Nephelometry/1

14. An antinuclear antibody (ANA) test was performed by using immunofluorescence microscopy assay (IFA), and a clinically significant pattern and titer were reported. Positive and negative controls performed as expected. However, the clinical evaluation of the patient was not consistent with the reported pattern. What is the most likely explanation for this situation?

- A. The clinical condition of the patient changed since the sample was tested
- B. The pattern of fluorescence was misinterpreted
- C. The control results were misinterpreted
- D. The wrong cell line was used for the test

Immunology/Apply principles of basic laboratory procedures/IFA/3

Answers to Questions 11–14

- 11. A** SPE should be performed initially to detect the presence of an abnormal Ig that demonstrates restricted electrophoretic mobility. A patient producing only monoclonal light chains may not show an abnormal serum finding because the light chains may be excreted in urine. A positive finding for either serum or urine should be followed by IFE on the positive specimen. This is required to confirm the presence of monoclonal Ig and to identify the heavy and light chain types.
- 12. B** A narrow dark band formed in both the lane containing anti- γ and anti- λ indicates the presence of a monoclonal IgG- λ . A diffuse dark band would indicate a polyclonal increase in IgG that often accompanies chronic inflammatory disorders, such as

systemic lupus erythematosus (SLE).

- 13. A** Rate nephelometry is used to measure the formation of small immune complexes as they are formed under conditions of antibody excess. The rate of increase in the photodetector output is measured within seconds or minutes, and the rate increases with increasing antigen concentration. Antigen concentration is determined by comparing the rate for the sample with that for standards by using an algorithm that compensates for nonlinearity. In endpoint nephelometry, reactions are read after equivalence. Immune complexes are of maximal size but may have a tendency to settle out of solution, thereby decreasing the amount of scatter.
- 14. B** In an IFA for antinuclear antibodies, the fluorescence pattern must be correlated correctly with the specificity of the antibodies. Both pathological and nonpathological antibodies can occur, and antibodies may be detected at a significant titer in a patient whose disease is inactive. Failure to correctly identify subcellular structures may result in misinterpretation of the antibody specificity or a false-positive result caused by nonspecific fluorescence.

15. What corrective action should be taken when a specific pattern cannot be identified in a specimen with a positive ANA IFA?

- A. Repeat the test using a larger volume of sample
- B. Call the physician
- C. Have another medical laboratory scientist read the slide
- D. Dilute the sample and retest

Immunology/Evaluate laboratory data to take corrective action according to predetermined criteria/IFA/3

16. Which statement best describes passive agglutination reactions used for serodiagnosis?

- A. Such agglutination reactions are more rapid because they are a single-step process
- B. Reactions require the addition of a second antibody
- C. Passive agglutination reactions require biphasic incubation
- D. Carrier particles for antigen, such as latex particles, are used

Immunology/Apply principles of basic laboratory procedures/Agglutination/1

17. What has happened in a titer, if tube Nos. 5 to 7 show a stronger reaction than tube Nos. 1 to 4?

- A. Prozone reaction
- B. Postzone reaction
- C. Equivalence reaction
- D. Poor technique

Immunology/Evaluate data to determine possible inconsistent results/Serological titration/3

18. What is the titer in tube No. 8 if tube No. 1 is undiluted and dilutions are doubled?

- A. 64
- B. 128
- C. 256
- D. 512

Immunology/Calculate/Serological titration/2

19. The directions for a slide agglutination test instruct that after mixing the patient's serum and antigen-coated latex particles, the slide must be rotated for 2 minutes. What

would happen if the slide were rotated for 10 minutes?

- A. Possible false-positive result
- B. Possible false-negative result
- C. No effect
- D. Depends on the amount of antibody present in the sample

Immunology/Apply principles of basic laboratory procedures/Agglutination/3

20. Which outcome indicates a negative result in a complement fixation test?

- A. Hemagglutination
- B. Absence of hemagglutination
- C. Hemolysis
- D. Absence of hemolysis

Immunology/Apply principles of basic laboratory procedures/Complement fixation/1

Answers to Questions 15–20

- 15. **D** An unexpected pattern may indicate the presence of more than one antibody. Diluting the sample may help to clearly show the antibody specificities, if they are found in different titers. If the pattern is still atypical, a new sample should be collected and the test repeated or the specimen should be tested by an alternative method, such as ELISA or multiplex.
- 16. **D** Most agglutination tests used in serology employ passive or indirect agglutination, where carrier particles are coated with the antigen. The carrier molecule is of sufficient size so that the reaction of the antigen with antibody results in the formation of a complex that is more easily visible.
- 17. **A** In tubes Nos.1 to 4, insufficient antigen is present to give a visible reaction because excess antibody has saturated all available antigen sites. After dilution of antibody, tubes Nos.1 to 4 have the equivalent concentrations of antigen and antibody to allow formation of visible complexes.
- 18. **B** The antibody titer is reciprocal of the highest dilution of serum giving a positive reaction. For doubling dilutions, each tube has one half the amount of serum as the previous tube. Because the first tube was undiluted (neat), the dilution in tube No. 8 is $(1/2)^7$ and the titer equals 2^7 or 128.
- 19. **A** Failure to follow directions, as in this case where the reaction was allowed to proceed beyond the recommended time, may result in a false-positive reading. Drying on the slide may lead to a possible erroneous positive reading.
- 20. **C** In complement fixation, hemolysis indicates a negative test result. The absence of hemolysis indicates that complement was fixed in an antigen–antibody reaction and, therefore, that the specific complement binding antibody was present in the patient's serum. Consequently, it was not available to react in the indicator system.

21. What effect does selecting the wrong gate have on the results when cells are counted by flow cytometry?

- A. No effect
- B. Failure to count the desired cell population

- C. Falsely elevated results
- D. Impossible to determine

Immunology/Apply principles of basic laboratory procedures/Flow cytometry/3

22. Which statement best describes immunophenotyping?

- A. Lineage determination by detecting antigens on the surface of the gated cells by using fluorescent antibodies
- B. Identification of cell maturity by using antibodies to detect antigens within the nucleus
- C. Identification and sorting of cells by front and side scatter of light from a laser
- D. Analysis of cells collected by flow cytometry by using traditional agglutination reactions

Immunology/Apply principles of basic laboratory procedures/Flow cytometry/1

23. A flow cytometry scattergram of a bone marrow sample shows a dense population of cells located in-between normal lymphoid and normal myeloid cells. What is the most likely explanation?

- A. The sample was improperly collected
- B. An abnormal cell population is present
- C. The laser optics are out of alignment
- D. The cells are most likely not leukocytes

Immunology/Apply principles of basic laboratory procedures/Flow cytometry/3

Answers to Questions 21–23

- 21. B** Gating is the step performed to select the correct cells to be counted. Failure to properly perform this procedure will result in problems in isolating and counting the desired cells. It is impossible to determine if the final result would be falsely elevated or falsely lowered by problems with gating.
- 22. A** *Immunophenotyping* refers to classification of cells (lineage and maturity assignment) with use of a panel of fluorescent-labeled antibodies directed against specific surface antigens on the cells. Antibodies are referred to by their CD number. Monoclonal antibodies having a common CD number do not necessarily bind to the same epitope but recognize the same antigen on the cell surface. Reactivity of the selected cells with a panel of antibodies differentiates lymphoid cells from myeloid cells and identifies the stage of cell maturation.
- 23. B** Lymphoid cells and myeloid cells display in predictable regions of the scatterplot because of their characteristic size and density. Lymphoid cells cause less forward scatter and side scatter from the laser compared with myeloid cells. A dense zone of cells in-between those regions is caused by the presence of a large number of abnormal cells, usually blasts. The lineage of the cells can be determined by immunophenotyping with a panel of fluorescent-labeled antibodies.

3.3

Infectious Diseases

1. Which serum antibody response usually characterizes the primary (early) stage of syphilis?

- A. Antibodies against syphilis are undetectable
- B. Detected 1 to 3 weeks after appearance of the primary chancre
- C. Detected in 50% of cases before the primary chancre disappears
- D. Detected within 2 weeks after infection

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/1

2. What substance is detected in the sample by the rapid plasma reagin (RPR) and Venereal Disease Research Laboratory (VDRL) tests for syphilis?

- A. Cardiolipin
- B. Anticardiolipin antibody (ACA)
- C. Anti-*Treponema pallidum* antibody
- D. *T. pallidum*

Immunology/Apply knowledge of fundamental biological characteristics/Syphilis/Testing/1

3. What type of antigen is used in the RPR card test?

- A. Live treponemal organisms
- B. Killed suspension of treponemal organisms
- C. Cardiolipin
- D. Tanned sheep cells

Immunology/Apply principles of basic laboratory procedures/Syphilis/Testing/1

4. Which of the following is the most sensitive test to detect congenital syphilis?

- A. VDRL
- B. RPR
- C. *T. pallidum* particle agglutination (TP-PA)
- D. Polymerase chain reaction (PCR)

Immunology/Apply principles of basic laboratory procedures/Syphilis/Testing/1

5. A biological false-positive reaction is *least* likely with which test for syphilis?

- A. VDRL
- B. TP-PA
- C. RPR
- D. All are equally likely to yield a false-positive result

Immunology/Apply principles of basic laboratory procedures/Syphilis/Testing/1

Answers to Questions 1–5

1. **B** During the primary stage of syphilis, about 90% of patients develop antibodies between 1 and 3 weeks after the appearance of the primary chancre.

2. **B** *Reagin* is the name for a nontreponemal antibody that appears in the serum of individuals with syphilis and is detected by the RPR and VDRL assays. Reagin reacts with cardiolipin, a lipid-rich extract of beef heart and other animal tissues.
3. **C** Cardiolipin is extracted from animal tissues, such as beef hearts, and attached to carbon particles. In the presence of reagin, the particles will agglutinate.
4. **D** PCR will amplify a very small amount of DNA from *T. pallidum* and allow for the detection of the organism in the infant. Antibody tests, such as VDRL and RPR, may detect maternal antibody only and do not indicate if the infant has been infected.
5. **B** The TP-PA test is more specific for *T. pallidum* compared with nontreponemal tests, such as the VDRL and RPR tests, and would be the least likely to yield a biological false-positive result. Nontreponemal tests have a biological false-positive rate of 1% to 10%, depending on the patient population tested. False-positive findings are caused commonly by infectious mononucleosis (IM), SLE, viral hepatitis, and human immunodeficiency virus (HIV) infection.

6. **A 12-year old girl has symptoms of fatigue and localized lymphadenopathy. Laboratory tests reveal peripheral blood lymphocytosis, positive RPR, and positive spot test for IM. What test should be performed next?**

- A. HIV screen
- B. VDRL**
- C. Epstein-Barr virus (EBV)–specific antigen test
- D. TP-PA test

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/3

7. **Which test is most likely to be positive in the tertiary stage of syphilis?**

- A. Treponemal-specific antibody
- B. RPR**
- C. VDRL
- D. Reagin screen test (RST)

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/3

8. **What is the most likely interpretation of the following syphilis serological results?**

RPR: reactive; TP-PA: nonreactive

- A. Neurosyphilis**
- B. Secondary syphilis
- C. Syphilis that has been successfully treated
- D. Biological false positive

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/2

9. **Which specimen is the sample of choice to evaluate latent or tertiary syphilis?**

- A. Serum sample
- B. Chancre fluid**
- C. Cerebrospinal fluid (CSF)
- D. Joint fluid

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/1

10. **Interpret the following quantitative RPR test results.**

RPR titer: weakly reactive—1:4; reactive—1:8 to 1:64

- A. Excess antibody, prozone effect
- B. Excess antigen, postzone effect
- C. Equivalence of antigen and antibody
- D. Impossible to interpret; testing error

Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/3

Answers to Questions 6–10

- 6. **D** The patient's symptoms are nonspecific and could be attributed to many potential causes. However, the patient's age, lymphocytosis, and serological results point to IM. The rapid spot test for antibodies seen in IM is highly specific. The EBV-specific antigen test is more sensitive but is unnecessary when the spot test is positive. HIV infection is uncommon at this age and is often associated with generalized lymphadenopathy and a normal or reduced total lymphocyte count. IM antibodies are commonly implicated as a cause of biological false-positive nontreponemal test results for syphilis. Therefore, a treponemal test for syphilis should be performed to document this phenomenon in this case.
- 7. **A** A treponemal-specific antibody test is more likely to be positive compared with a nontreponemal test in the tertiary stage of syphilis. In some cases, systemic lesions have subsided by the tertiary stage, and the nontreponemal tests become seronegative. Although the treponemal-specific antibody test is the most sensitive test for tertiary syphilis, it will be positive in both treated and untreated cases.
- 8. **D** A positive reaction with nontreponemal antigen and a negative reaction with a treponemal antigen is most likely caused by a biological false-positive nontreponemal test result.
- 9. **C** Latent syphilis usually begins after the second year of untreated infection. In some cases, the serological tests become negative. However, if neurosyphilis is present, CSF serology will be positive and the CSF will display increased protein and pleocytosis characteristic of central nervous system infection.
- 10. **A** This patient may be in the secondary stage of syphilis and is producing large amounts of antibody to *T. pallidum* sufficient to cause a prozone reaction as a result of antibody excess in the test. The test became strongly reactive only after the antibody was diluted.

11. Tests to identify infection with HIV fall into which three general classification types of tests?

- A. Tissue culture, antigen, and antibody tests
- B. Tests for antigens, antibodies, and nucleic acid
- C. DNA probe, DNA amplification, and Western blot tests
- D. ELISA, Western blot, and Southern blot tests

Immunology/Apply principles of basic laboratory procedures/HIV/Testing/1

12. Which tests are considered screening tests for HIV?

- A. ELISA, chemiluminescent, and rapid antibody tests

- B. IFA, Western blot, radioimmunoprecipitation assay
- C. Culture, antigen capture assay, DNA amplification
- D. Reverse transcriptase and messenger RNA (mRNA) assay

Immunology/Apply principles of basic laboratory procedures/HIV/Testing/1

13. Which tests are the recommended confirmatory tests for HIV?

- A. ELISA and rapid antibody tests
- B. HIV-1,2 antibody differentiation assays, and qualitative PCR test
- C. Culture, antigen capture assay, quantitative PCR
- D. Reverse transcriptase and mRNA assay

Immunology/Apply principles of basic laboratory procedures/HIV/Testing/1

14. How do fourth- and fifth-generation HIV tests reduce the time from infection to the test becoming positive?

- A. They are PCR tests detecting viral RNA
- B. They detect p24 antigen in addition to HIV antibody
- C. They detect proviral DNA
- D. They detect antibodies to more antigens than earlier generations of HIV tests

Immunology/Evaluate laboratory data to recognize health and disease states/HIV/4th gen tests/2

15. A woman who has had five pregnancies subsequently tests positive for HIV on a fourth-generation assay and is negative on an HIV-1,2 differentiation assay and a follow-up molecular assay. The initial reactivity may be caused by:

- A. Possible cross-reaction with herpes or EBV antibodies
- B. Interference from medication
- C. Cross-reacting antibodies elicited during pregnancy
- D. Possible technical error; a repeat specimen should be requested

Immunology/Evaluate laboratory data to recognize health and disease states/HIV/algorithm interpretation/3

16. Interpret the following results for HIV testing:

Fourth-generation ELISA: positive; repeat ELISA: positive; HIV 1,2 antibody differentiation assay: negative; qualitative HIV RNA rtPCR assay: positive

- A. False-positive fourth-generation assay
- B. False-negative antibody differentiation assay
- C. Indeterminate; further testing indicated
- D. HIV p24 antigen detected on fourth-generation ELISA

Immunology/Evaluate laboratory data to recognize health and disease states/HIV/Testing/3

Answers to Questions 11–16

- 11. B** The fourth- and fifth-generation HIV assays detect both antibodies to HIV and the HIV p24 antigen. Molecular assays can be used to resolve discrepant screening results or to confirm results, as well as to quantitate the amount of virus present.
- 12. A** The fourth- and fifth-generation HIV assays detect both antibody and the p24 antigen. These assays are available in ELISA, automated chemiluminescent systems, and

multiplex systems and in a rapid card format.

- 13. **B** The current HIV testing algorithm begins with a screening assay, followed by an HIV-1,2 antibody differentiation assay (HIV-1,2 supplemental assay) and, if those results are discordant, an HIV qualitative PCR assay is performed. Western blot is not included in this algorithm.
- 14. **B** Including the p24 antigen in the fourth- and fifth-generation tests allows for the detection of HIV infection approximately 1 week earlier compared with the third-generation antibody-only assays and up to 3 weeks earlier compared with Western blot.
- 15. **C** Pregnancy is a common cause of false-positive HIV screening results.
- 16. **D** The fourth-generation HIV assay detects antibody and the p24 antigen but does not differentiate between those results. In this case, the antibody-confirming test is negative, suggesting the initial reactive fourth-generation test result is either a false-positive one or is caused by the presence of p24 in the specimen. The positive molecular assay confirms the presence of the virus in the specimen. This usually occurs in early infection, prior to antibody being produced.

17. What is the most likely explanation when antibody tests for HIV are negative but the PCR test is positive?

- A. Probably not HIV infection
- B. Patient is in the “window phase” before antibody production
- C. Tests were performed incorrectly
- D. Clinical signs may be misinterpreted

Immunology/Correlate laboratory data with physiological processes/HIV/Testing/3

18. What criteria constitute the classification system for HIV infection?

- A. CD4-positive T-cell count and clinical symptoms
- B. Clinical symptoms, condition, duration, and strength of reactivity on a fourth-generation HIV test
- C. Presence or absence of lymphadenopathy
- D. Strong fourth-generation HIV test reactivity and CD8-positive T-cell count

Immunology/Apply knowledge of fundamental biological characteristics/HIV/2

19. What is the main difficulty associated with the development of an HIV vaccine?

- A. The virus has been difficult to culture; antigen extraction and concentration are extremely laborious
- B. Human trials cannot be performed
- C. Different strains of the virus are genetically diverse
- D. Anti-idiotypic antibodies cannot be developed

Immunology/Apply principles of basic immunology/HIV/Vaccines/2

20. Which CD4:CD8 ratio is most likely in a patient with AIDS?

- A. 2:1
- B. 3:1
- C. 2:3
- D. 1:3

Immunology/Correlate laboratory data with physiological processes/HIV/Testing/2

Answers to Questions 17–20

- 17. B** In early seroconversion, patients may not be making antibodies in sufficient amounts to be detected by antibody tests. The period between infection with HIV and the appearance of detectable antibodies is called the *window phase*. This period has been reduced to a few weeks by antibody and antigen-detecting fourth- and fifth-generation assays, and an algorithm that includes PCR testing.
- 18. A** The classification (not diagnostic) system for HIV infection is based on a combination of CD4-positive T-cell count (helper T cells) and various categories of clinical symptoms. Classification is important in determining treatment options and the progression of the disease.
- 19. C** Vaccine development has been difficult primarily because of the genetic diversity among different strains of the virus, and new strains are constantly emerging. HIV-1 can be divided into two main subtypes designated M (for main) and O (for outlier). The M group is further divided into nine subgroups, designated A through J (there is no E subgroup), based on differences in the nucleotide sequence of the *gag* gene. Two remaining subtypes are designated N (non-M and non-O) and P (a subtype related to SIVgor). A vaccine that is effective for all of the subgroups of HIV-1 has yet to be developed.
- 20. D** An inverted CD4:CD8 ratio (less than 1.0) is a common finding in a patient with AIDS. The Centers for Disease Control and Prevention (CDC) requires a CD4-positive (helper T) cell count of less than 200/ μ L or 14% in the absence of an AIDS-defining illness (e.g., *Pneumocystis carinii* pneumonia) in the case surveillance definition of AIDS.

21. What is the advantage of fourth-generation rapid HIV tests over earlier rapid HIV tests?

- A. They use recombinant antigens
- B. They detect multiple strains of HIV
- C. They detect p24 antigen
- D. They are quantitative

Immunology/Apply principles of basic laboratory procedures/HIV/Testing/2

22. Which method is used to test for HIV infection in infants who are born to HIV-positive mothers?

- A. ELISA
- B. Western blot test
- C. PCR test
- D. Viral culture

Immunology/Apply principles of special procedures/HIV/1

23. What is the most likely cause when a fourth-generation HIV assay is positive for all controls and samples?

- A. Improper pipetting
- B. Improper washing
- C. Improper addition of sample

D. Improper reading

Immunology/Evaluate laboratory data to recognize problems/HIV/Testing/3

24. What constitutes a diagnosis of viral hepatitis?

- A. Abnormal test results for liver enzymes
- B. Clinical signs and symptoms
- C. Positive results for hepatitis markers
- D. All of these options

Immunology/Evaluate laboratory data to recognize health and disease states/Hepatitis/Testing/2

25. Which of the following statements regarding infection with hepatitis D virus (HDV) is true?

- A. Occurs in patients with HIV infection
- B. Does not progress to chronic hepatitis
- C. Occurs in patients with hepatitis B virus (HBV) infection
- D. Is not spread through blood or sexual contact

Immunology/Apply knowledge of fundamental biological characteristics/Hepatitis/1

26. All of the following hepatitis viruses are spread through blood or blood products except:

- A. Hepatitis A virus (HAV)
- B. HBV
- C. HCV
- D. HDV

Immunology/Apply knowledge of fundamental biological characteristics/Hepatitis/1

Answers to Questions 21–26

- 21. C Both third-generation and fourth-generation rapid tests for HIV use recombinant and synthetic HIV antigens conjugated to a solid phase. The multivalent nature of these tests allows for detection of less common subgroups of HIV-1 and simultaneous detection of both HIV-1 and HIV-2. However, the fourth-generation assays also use solid-phase antibodies to p24 antigen to detect its presence. Because p24 antigen appears before antibodies to HIV, fourth-generation tests can detect infection 4 to 7 days earlier compared with tests based on antibody detection alone.
- 22. C Fourth- and fifth-generation ELISA and chemiluminescent assays reflect the presence of maternal antibody. The PCR test uses small amounts of blood and does not rely on the antibody response. PCR amplifies small amounts of viral nucleic acid and can detect less than 20 copies of viral RNA per milliliter of plasma. These qualities make PCR ideal for the testing of infants. Nucleic acid methods for HIV RNA include both qualitative (for diagnosis) and quantitative (for monitoring) reverse-transcriptase real-time PCR (RT-PCR) assays.
- 23. B Improper washing may not remove unbound, enzyme-conjugated antihuman Ig, and every sample may appear positive.
- 24. D To diagnose a case of hepatitis, the physician must consider clinical signs as well as the results of laboratory tests that measure liver enzymes and hepatitis markers.
- 25. C HDV is an RNA virus that requires the surface antigen or envelope of the HBV for

entry into the hepatocyte. Consequently, HDV can infect only patients who are coinfecting with hepatitis B.

- 26. A** HAV is spread through the fecal–oral route and is the cause of infectious hepatitis. HAV has a shorter incubation period (2–7 weeks) than HBV (1–6 months). Epidemics of HAV can occur, especially when food and water become contaminated with raw sewage. Hepatitis E virus is also spread via the oral–fecal route and, like HAV, has a short incubation period.

27. Which hepatitis B marker is the best indicator of early acute infection?

- A. Hepatitis B surface antigen (HBsAg)
- B. Hepatitis B e-antigen (HBeAg)
- C. Hepatitis B core antibody (anti-HBc)
- D. Hepatitis B surface antibody (anti-HBs)

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/2

28. Which is the first antibody detected in serum after infection with HBV?

- A. Anti-HBs
- B. Anti-HBc IgM
- C. Anti-HBe
- D. All are detectable at the same time

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/2

29. Which antibody persists in low-level carriers of HBV?

- A. IgM anti-HBc
- B. IgG anti-HBc
- C. IgM anti-HBe
- D. IgG anti-HBs

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/2

30. What is the most likely explanation when a patient has clinical signs of viral hepatitis but tests negative for HAV IgM, HBsAg, and HCV antibody?

- A. Tests were performed improperly
- B. The patient does not have hepatitis
- C. The patient may be in the “core window”
- D. Clinical evaluation was performed improperly

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/3

31. Which hepatitis B markers should be performed on blood products?

- A. HBsAg and anti-HBc
- B. Anti-HBs and anti-HBc
- C. HBeAg and HBcAg
- D. Anti-HBs and HBeAg

Immunology/Apply principles of laboratory operations/Hepatitis/Testing/1

32. Which hepatitis antibody confers immunity against reinfection with HBV?

- A. Anti-HBc IgM
- B. Anti-HBc IgG
- C. Anti-HBe
- D. Anti-HBs

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/1

Answers to Questions 27–32

27. **A** HBsAg is the first marker to appear in HBV infection. It is usually detected within 4 weeks of exposure (prior to the rise in transaminases) and persists for about 3 months after serum enzyme levels return to normal.
28. **B** Antibody to the hepatitis B core antigen (anti-HBc) is the first detectable hepatitis B antibody. It persists in serum for years after infection and is found in the serum of asymptomatic carriers of HBV. Because levels of total anti-HBc are high after recovery, IgM anti-HBc is a more useful marker for acute infection. Both anti-HBc and anti-HBs can persist for life, but only anti-HBs is considered protective.
29. **B** IgG anti-HBc can be detected in carriers who are HBsAg and anti-HBs negative. These persons are hepatitis B DNA positive also and, thus, are presumed infective, even though the level of HBsAg is too low to detect. No specific B core IgG test is available, however. This patient would be positive in the anti-B core total antibody assay and negative in the anti-HB core IgM test.
30. **C** The patient may be in the “core window,” the period of HBV infection when both the surface antigen and surface antibody are undetectable. The IgM anti-HBc and the anti-HBc total antibody assays, along with the hepatitis B DNA PCR assay would be the only detectable markers in the serum of a patient in the core window phase of HBV infection.
31. **A** Blood products are tested for HBsAg, an early indicator of infection, and anti-HBc, a marker that may persist for life. Following recovery from HBV infection, some patients demonstrate negative serology for HBsAg and anti-HBs but are positive for anti-HBc. Such patients are considered infective.
32. **D** Anti-HBs appears later in infection compared with anti-HBc and is used as a marker for immunity after infection or vaccination, rather than for diagnosis of current infection.

33. Which test, other than serological markers, is most consistently elevated in viral hepatitis?

- A. Antinuclear antibodies
- B. Alanine aminotransferase (ALT)
- C. Absolute lymphocyte count
- D. Lactate dehydrogenase

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/1

34. If only anti-HBs is positive, which of the following can be ruled out?

- A. HBV vaccination
- B. Distant past infection with HBV
- C. Hepatitis B immune globulin (HBIG) injection
- D. Chronic HBV infection

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/2

35. Interpret the following results for EBV infection: IgG and IgM antibodies to viral capsid antigen (VCA) are positive.

- A. Infection in the past
- B. Infection with a mutual enhancer virus, such as HIV
- C. Current infection
- D. Impossible to interpret; need more information

Immunology/Correlate laboratory data with physiological processes/EBV/Testing/3

36. Rapid mono tests use latex particles coated with which of the following?

- A. Guinea pig antigen
- B. Beef proteins
- C. Horse proteins
- D. Sheep proteins

Immunology/Apply principles of basic laboratory procedures/IM/Testing/1

Answers to Questions 33–36

- 33. B** ALT is a liver enzyme and may be increased in hepatic disease. Highest levels occur in acute viral hepatitis, reaching 20 to 50 times the upper limit of normal.
- 34. D** Persons with chronic HBV infection show a positive test result for anti-HBc (IgG or total) and HBsAg but not anti-HBs. Patients with active chronic hepatitis have not become immune to the virus.
- 35. C** IgM and IgG antibodies to VCA are found in a current infection with EBV. The IgG antibody may persist for life, but the IgM anti-VCA disappears within 4 months after the infection resolves.
- 36. B** Rapid mono tests detect a heterophile antibody directed against beef proteins. Although these antibodies may also react with horse or sheep red blood cells (RBCs), those proteins are not used in these tests.

37. Blood products are tested for which virus before being transfused to newborns?

- A. EBV
- B. Human T-lymphotropic virus II (HTLV-II)
- C. CMV
- D. HDV

Immunology/Apply principles of laboratory operations/Virus testing/1

38. What is the endpoint for the antistreptolysin O (ASO) latex agglutination assay?

- A. Highest serum dilution that shows no agglutination
- B. Highest serum dilution that shows agglutination
- C. Lowest serum dilution that shows agglutination
- D. Lowest serum dilution that shows no agglutination

Immunology/Apply principles of basic laboratory procedures/ASO/Interpretation/1

Answers to Questions 37–38

- 37. C** CMV can be life threatening if transmitted to a newborn through a blood product. HTLV-II is a rare virus, which like HIV, is a T-cell tropic RNA retrovirus. The virus has been associated with hairy cell leukemia, but this is not a consistent finding.

38. B The latex test for ASO includes latex particles coated with streptolysin O. Serial dilutions are prepared and the highest dilution showing agglutination is the endpoint.

39. A streptozyme test was performed, but the result was negative, even though the patient showed clinical signs of a streptococcal throat infection. What should be done next?

- A. Either ASO or anti-deoxyribonuclease B (anti-DNase B) test
- B. Another streptozyme test using diluted serum
- C. Antihyaluronidase test
- D. Wait for 3 to 5 days and repeat the streptozyme test

Immunology/Evaluate laboratory data to recognize health and disease states/ASO/Testing/3

40. Rapid assays for influenza that utilize specimens obtained from nasopharyngeal swabs detect:

- A. IgM anti-influenza
- B. IgA anti-influenza
- C. IgA–influenza antigen immune complexes
- D. Influenza nucleoprotein antigens

Immunology/Apply principles of basic laboratory procedures/Virus testing/2

41. How can interfering cold agglutinins be removed from a test sample?

- A. Centrifuge the serum and remove the top layer
- B. Incubate the clot at 1°C to 4°C for several hours and then remove the serum
- C. Incubate the serum at 56°C in a water bath for 30 minutes
- D. Use an anticoagulated sample

Immunology/Apply principles of special procedures/Cold agglutinins/Testing/2

42. All tubes (dilutions) except the negative control are positive for cold agglutinins. This indicates:

- A. Contaminated RBCs
- B. A rare antibody against RBC antigens
- C. The sample was stored at 4°C prior to separating serum and cells
- D. Further serial dilution is necessary

Immunology/Select course of action/Cold agglutinins/Testing/3

43. All positive cold agglutinin tubes remain positive after 37°C incubation except the positive control. What is the most likely explanation for this situation?

- A. High-titer cold agglutinins
- B. Contamination of the test system
- C. Antibody other than cold agglutinins
- D. Faulty water bath

Immunology/Evaluate laboratory data to determine possible inconsistent results/Cold agglutinins/Testing/3

44. Which increase in antibody titer (dilution) best indicates an acute infection?

- A. From 1:2 to 1:8
- B. From 1:4 to 1:16
- C. From 1:16 to 1:256
- D. From 1:64 to 1:128

Immunology/Correlate laboratory data with physiological processes/Antibody titers/1

Answers to Questions 39–44

39. **A** The streptozyme test is used for screening and contains several of the antigens associated with streptococcal products. Because some patients produce an antibody response to a limited number of streptococcal products, no single test is sufficiently sensitive to rule out infection. Clinical sensitivity is increased by performing additional tests when initial results are negative. The streptozyme test generally yields more false-positive and false-negative results compared with the ASO and anti-DNase B tests. A positive result occurs in a smaller number of patients with recent streptococcal infections in the antihyaluronidase test compared with the ASO and anti-DNase B tests.
40. **D** The rapid influenza assays are antigen detection methods. They are designed to detect early infection, before antibody is produced.
41. **B** Cold agglutinins will attach to autologous RBCs if incubated at 1°C to 4°C. The absorbed serum will be free of cold agglutinins.
42. **D** Cold agglutinins may be measured in patients who have cold agglutinin disease, that is, cold autoimmune hemolytic anemia. In such cases, titers can be as high as 10^6 . If all tubes (dilutions) for cold agglutinins are positive, except the negative control, then a high titer of cold agglutinins is present in the sample. Further serial dilutions should be performed.
43. **C** Cold agglutinins do not remain reactive above 30°C, and agglutination must disperse after incubation at 37°C. The most likely explanation when agglutination remains after 37°C incubation is that a warm alloantibody or autoantibody is present.
44. **C** A fourfold (two-tube) or greater increase in antibody titer is usually indicative of an acute infection. Although answers A and B show a fourfold rise in titer, answer C shows a 16-fold rise in titer and is the most definitive. In most serological tests, a single high titer is insufficient evidence of acute infection unless specific IgM antibodies are measured because age, individual variation, immunologic status, and history of previous exposure (or vaccination) cause a wide variation in normal serum antibody titers.

45. Which of the following positive antibody tests may be an indication of recent vaccination or early primary infection for rubella in a patient with no clinical symptoms?

- A. Only IgG antibodies positive
- B. Only IgM antibodies positive
- C. Both IgG and IgM antibodies positive
- D. Fourfold rise in titer for IgG antibodies

Immunology/Apply principles of basic laboratory procedures/Rubella/Testing/2

46. Why is laboratory diagnosis difficult in cases of Lyme disease?

- A. Clinical response may not be apparent upon initial infection; IgM antibody may not be detected until 3 to 6 weeks after the infection
- B. Laboratory tests may be designed to detect whole *Borrelia burgdorferi*, not flagellar antigen found early in infection

- C. Most laboratory tests are technically demanding and lack specificity
- D. Antibodies formed initially to *B. burgdorferi* may cross react in antigen tests for autoimmune diseases

Immunology/Correlate clinical signs with laboratory procedures/Lyme disease/Testing/2

47. Serological tests for which disease may give a false-positive result if the patient has Lyme disease?

- A. HIV
- B. Syphilis
- C. EBV
- D. Hepatitis C

Immunology/Evaluate laboratory data to determine possible inconsistent results/Lyme disease/Testing/3

48. In monitoring a patient with HIV infection, which parameter may be expected to be the most sensitive indicator of the effectiveness of antiretroviral treatment?

- A. HIV antibody titer
- B. CD4:CD8 ratio
- C. HIV viral load
- D. Absolute total T-cell count

Immunology/Correlate clinical and laboratory data/HIV/2

49. A renal transplant recipient is found to have a rising creatinine level and reduced urine output. The physician orders a “urine PCR” assay. When you call to find out what organism the physician wants to identify, you are told:

- A. HCV
- B. *Legionella pneumophila*
- C. EBV
- D. BK virus

Immunology/Apply knowledge of fundamental biological characteristics/Transplant/Virus/2

50. A newborn is to be tested for vertically transmitted HIV infection. Which of the following tests is most useful?

- A. HIV PCR
- B. CD4 count
- C. Rapid HIV antibody test
- D. HIV IgM antibody test

Immunology/Select test/Neonatal HIV/2 Answers to Questions 45–50

Answers to Questions 45–50

- 45. B** If only IgM antibodies are positive, this result indicates recent vaccination or early primary infection.
- 46. A** Lyme disease is caused by *B. burgdorferi*, a spirochete, and typical clinical symptoms, such as rash or erythema chronicum migrans, may be absent in some infected individuals. Additionally, IgM antibody is not detectable by laboratory tests until 3 to 6 weeks after a tick bite, and IgG antibody develops later.
- 47. B** Lyme disease is caused by a spirochete, and positive results may occur with some

specific treponemal antibody tests for syphilis.

48. **C** The HIV viral load will rise or fall in response to treatment more quickly compared with any of the other listed parameters. The absolute CD4 count is also an indicator of treatment effectiveness and is used in resource-poor areas that might not have facilities for molecular testing. Note, however, that the absolute CD4 count is not one of the choices.
49. **D** BK virus is a polyoma virus that can cause renal and urinary tract infections. The virus is an opportunistic pathogen and has become a well-recognized cause of poor renal function in kidney transplant recipients. Antibody testing is not practical or useful for this infection. The principal diagnostic assays are urinary cytology, and specific BK virus PCR testing in urine and serum. Although *L. pneumophila* can be diagnosed through a urinary antigen assay, that organism is not a primary cause of renal insufficiency in transplant recipients.
50. **A** Neonatal HIV diagnosis is performed by screening for the presence of the virus. The current antibody tests are either an IgG-specific assay or an IgG/IgM combination assay. Thus, an infant whose mother is HIV positive will also be positive in the HIV antibody assay. Although the CD4 count may be a useful assay to determine disease activity, there are many causes of reduced CD4 numbers, so this assay should not be used to diagnose HIV infection.

51. Which of the following fungal organisms is best diagnosed by an antigen detection test as opposed to an antibody detection assay?

- A. *Histoplasma*
- B. *Cryptococcus*
- C. *Candida*
- D. *Aspergillus*

Immunology/Apply knowledge of special procedures/Fungal testing/2

52. Your cytology laboratory refers a Papanicolaou smear specimen to you for an assay designed to detect the presence of a virus associated with cervical cancer. You perform:

- A. An ELISA for anti-human simplex virus 2 (anti-HSV-2) antibodies
- B. A molecular assay for HSV-2
- C. An ELISA for human papilloma virus (HPV) antibodies
- D. A molecular assay for HPV

Immunology/Select course of action/Virus testing/Methods/3

53. An immunosuppressed patient has unexplained anemia. The physician suspects a parvovirus B19 infection. The parvovirus IgM test result is negative. The next course of action is to tell the physician that:

- A. The patient does not have parvovirus
- B. A convalescent specimen is recommended in 4 weeks to determine if a fourfold rise in titer has occurred
- C. A parvovirus PCR is recommended
- D. A recent transfusion for the patient's anemia may have resulted in a false-negative result and the patient should be retested in 4 weeks

Immunology/Select course of action/Virus testing/Parvovirus/3

Answers to Questions 51–53

- 51. B** The *Cryptococcus* antibody response is not a reliable indicator of a current infection; thus, an antigen assay is normally used to monitor the disease. The antigen assay may be used for serum or CSF and will decline in response to treatment much faster than a traditional antibody test. A urinary antigen test is available for histoplasmosis, and a serum galactomannan assay is available for *Aspergillus*. Those two assays perform better than antibody detection. No antigen test is available for *Candida*, and thus, antibody detection is the best serological procedure for this organism.
- 52. D** Cervical cell atypia and cervical cancer are associated with specific high-risk serotypes of HPV infections. Although HPV antibody assays are available, they are not serotype specific, nor do they relate to disease activity. Thus, molecular probe assays are the tests of choice to detect high-risk HPV infection. Although HSV-2 is associated with genital herpesvirus, that virus has not been shown to cause cervical cancer.
- 53. C** A negative IgM assay rarely rules out an infection. Although a convalescent specimen may be useful in many cases, in an immunosuppressed patient, the convalescent specimen may remain negative in the presence of an infection. Thus, a parvovirus PCR test is the preferred choice in this case. A false-negative result could be caused by multiple whole blood or plasma transfusions, but retesting for antibody a month later would not be beneficial to the patient.

3.4

Autoimmune Diseases

1. What is a general definition for autoimmunity?

- A. Increase of tolerance to self-antigens
- B. Loss of tolerance to self-antigens
- C. Increase in clonal deletion of mutant cells
- D. Manifestation of immunosuppression

Immunology/Apply knowledge of fundamental biological characteristics/Autoimmunity/Definitions/1

2. An ANA test is performed on a specimen from a 55-year-old woman who has unexplained joint pain. The IFA result shows a titer of 40 and a homogeneous pattern. The appropriate follow-up for this patient is:

- A. Anti-DNA assay
- B. Extractable nuclear antigen (ENA) testing
- C. Retest ANA in 3 to 6 months
- D. CH50 complement assay

Immunology/Correlate laboratory data with physiological processes/IF/2

3. Which disease is likely to show a rim (peripheral) pattern in an immunofluorescence (IF) microscopy test for ANA?

- A. Mixed connective tissue disease (MCTD)
- B. RA
- C. SLE
- D. Scleroderma

Immunology/Correlate laboratory data with physiological processes/IF/2

4. A patient's specimen is strongly positive in an ANA ELISA. Which of the following would *not* be an appropriate follow-up to this result?

- A. IFA on human epithelial type 2 (HEp-2) cells
- B. Specific ENA ELISA tests
- C. Specific anti-DNA ELISA
- D. Rheumatoid factor (RF) assay

Immunology/Select tests/ANA Confirmation/2

5. What type of antibodies is represented by the homogeneous pattern in the IFA for ANAs?

- A. Antihistone antibodies
- B. Anticentromere antibodies
- C. Anti-ENA (anti-Smith [anti-Sm] and anti-ribonucleoprotein [anti-RNP]) antibodies
- D. Anti-RNA antibodies

Immunology/Correlate laboratory data with physiological processes/IF/1

6. What disease is indicated by a high titer of anti-Sm antibody?

- A. MCTD

- B. RA
- C. SLE
- D. Scleroderma

Immunology/Correlate laboratory data with physiological processes/IF/2

Answers to Questions 1–6

1. **B** Autoimmunity is a loss of tolerance to self-antigens and the subsequent formation of autoantibodies.
2. **C** Approximately 25% of women in this age range may have low titer–positive ANA assays with no demonstrable connective tissue disease. A patient with anti-DNA–positive SLE would be expected to have a much higher titer (greater than 160) in an IFA. A similar titer would be expected for an ENA positive specimen, although the pattern would be speckled. Complement testing would not be indicated with this low titer in a 55-year-old female.
3. **C** The rim or peripheral pattern seen in indirect immunofluorescence techniques is most commonly found in cases of active SLE. The responsible autoantibody is highly correlated to anti–double-stranded DNA (anti-dsDNA).
4. **D** The ANA ELISA is a screening assay. A positive result may be followed up by more specific antibody ELISA tests or an ANA IFA to determine pattern and titer. The ANA ELISA does not screen for RF.
5. **A** Antihistone antibodies (and also anti-DNA antibodies) cause the solid or homogeneous pattern, which is commonly found in patients with SLE, RA, MCTD, and Sjögren syndrome. Antibodies to the centromere of chromosomes is a marker for the CREST (calcinosis, Raynaud phenomenon, esophageal dysfunction, sclerodactyly, and telangiectasia) form of systemic sclerosis.
6. **C** High titer anti-Sm is indicative of SLE. Anti-Sm is an antibody against saline ENAs. and causes a speckled pattern of immunofluorescence.

7. Which disease is *least* likely when a nucleolar pattern occurs in an IFA for ANAs?

- A. MCTD
- B. Sjögren syndrome
- C. SLE
- D. Scleroderma

Immunology/Correlate laboratory data with physiological processes/IF/2

8. What antibodies are represented by the nucleolar pattern in the IFA for ANAs?

- A. Antihistone antibodies
- B. Anti-dsDNA antibodies
- C. Anti-ENA (anti-Sm and anti-RNP) antibodies
- D. Anti-RNA antibodies

Immunology/Correlate laboratory data with physiological processes/IF/1

9. Which test would *best* distinguish between SLE and MCTD?

- A. Multiplex or ELISA test for anti-Sm and anti-RNP
- B. IFA using *Crithidia* as substrate

- C. Slide agglutination testing
- D. Laboratory tests cannot distinguish between these disorders

Immunology/Evaluate laboratory data to recognize and report the need for additional testing/Autoimmune/Testing/3

- 10. An ANA test on HEP-2 cells shows nucleolar staining in interphase cells and dense chromatin staining in mitotic cells. The most likely cause of this staining pattern is:**
- A. Antifibrillar antibody
 - B. Anti-ribosomal P antibody
 - C. A serum with nucleolar and homogeneous patterns
 - D. Technical artifact

Immunology/Correlate laboratory data with physiological processes/IF/1

- 11. Which immunofluorescence pattern indicates the need for ENA testing by multiplex, line blots, or ELISA assays?**
- A. Homogeneous or solid
 - B. Peripheral or rim
 - C. Speckled
 - D. Nucleolar

Immunology/Evaluate laboratory data to recognize and report the need for additional testing/Autoimmune/Testing/3

- 12. Which of the following is used in rapid slide tests for detection of RFs?**
- A. Whole IgM molecules
 - B. Fc portion of the IgG molecule
 - C. Fab portion of the IgG molecule
 - D. Fc portion of the IgM molecule

Immunology/Apply knowledge of fundamental biological characteristics/RA/Testing/1

Answers to Questions 7–12

- 7. A** All of the diseases except MCTD may cause a nucleolar pattern of immunofluorescence. Nucleolar fluorescence is caused by anti-RNA antibodies and is seen in about 50% of patients with scleroderma.
- 8. D** Anti-RNA antibodies are represented by the nucleolar pattern. This pattern may be seen in most systemic autoimmune diseases and is especially common in patients with scleroderma. Anti-RNA and anti-Sm are not usually found in patients with MCTD. This is a syndrome involving aspects of SLE, RA, scleroderma, and polymyositis. The immunofluorescence pattern most often seen in MCTD is the speckled pattern caused by anti-RNP.
- 9. A** Line blots, multiplex, and ELISA assays, using purified or recombinant antigens, are available for differentiating anti-RNP from anti-Sm. Anti-Sm with or without anti-RNP is found in approximately one third of patients with SLE. Anti-RNP in the absence of anti-Sm is found in over 95% of patients with MCTD.
- 10. A** Antifibrillar antibody has this appearance. Ribosomal P antibody has nucleolar staining and a background homogeneous and cytoplasmic stain. A combination nucleolar/homogeneous specimen will also show homogeneous staining in the

interphase cells. This pattern is not seen in typical technical artifacts.

- 11. C** A speckled pattern is often caused by the presence of antibodies against the ENAs, such as Sm, RNP, SSA, and SSB. Homogeneous and rim patterns suggest antibodies to dsDNA. The homogeneous pattern may also be seen with antibodies to deoxyribonuclear protein, which is not an ENA. Nucleolar patterns often indicate antibodies to RNA or fibrillarin.
- 12. B** RFs react with the Fc portion of the IgG molecule and are usually IgM. This is the basis of rapid agglutination tests for RA. Particles of latex or cells are coated with IgG. Addition of serum containing RF results in visible agglutination.

13. Which of the following methods is *least* likely to give a definitive result for the diagnosis of RA?

- A. Nephelometric measurement of anti-IgG
- B. Agglutination testing for RF
- C. Anti-CCP
- D. IFA for ANAs

Immunology/Select routine laboratory procedures/Autoimmune/RA/Testing/1

14. Which disease might be indicated by antibodies to smooth muscle?

- A. Atrophic gastritis
- B. Autoimmune hepatitis
- C. Myasthenia gravis
- D. Sjögren syndrome

Immunology/Apply knowledge of fundamental biological characteristics/Autoimmune/Testing/1

15. Antibodies to thyroid peroxidase may appear in which of the following diseases?

- A. Graves disease and Hashimoto thyroiditis
- B. Myasthenia gravis
- C. Granulomatous thyroid disease
- D. Addison disease

Immunology/Select routine laboratory procedures/Autoimmune/Testing/1

16. What is the main use of laboratory tests to detect antibodies to islet cells and insulin in cases of insulin-dependent diabetes mellitus (IDDM)?

- A. To regulate levels of injected insulin
- B. To diagnose IDDM
- C. To rule out the presence of other autoimmune diseases
- D. To screen susceptible individuals prior to destruction of β -cells

Immunology/Select routine laboratory procedures/Autoimmune/IDDM/Testing/1

17. A patient presents with clinical symptoms of celiac disease. Tests for anti-tissue transglutaminase and antigliadin antibodies are negative. Which of the following tests should be ordered?

- A. IgG level
- B. HLA DQ2 and DQ8 typing
- C. HLA DR3 and DR7 typing
- D. IgM level

Immunology/Select routine laboratory procedures/Autoimmune/Celiac disease/Testing/2

Answers to Questions 13–17

- 13. D** Patients with RA often show a homogeneous pattern of fluorescence in tests for ANAs. However, this pattern is seen in a wide range of systemic autoimmune diseases and in many normal persons at a titer below 10. The first two methods listed may be used to identify anti-IgG, which is one laboratory criterion used to establish a diagnosis of RA. Anti-CCP is an additional laboratory criterion used in the RA diagnostic algorithm.
- 14. B** Antibodies to smooth muscle are found in the serum of up to 70% of patients with active chronic hepatitis and up to 50% of patients with primary biliary cirrhosis.
- 15. A** Antibodies to thyroid peroxidase may be detected in both Graves disease (hyperthyroidism) and Hashimoto thyroiditis (hypothyroidism). If a positive result is found to thyroid peroxidase, thyroxine levels and clinical presentation can be used to distinguish between the two diseases.
- 16. D** Fasting hyperglycemia and hemoglobin A_{1c} levels are the primary findings used to diagnose IDDM. For individuals with an inherited susceptibility to the development of IDDM, laboratory tests for the detection of antibodies to islet cells and insulin may help to initiate early treatment before complete destruction of β -cells.
- 17. B** Although antibodies to tissue transglutaminase and gliadin are often found in celiac disease, their combined sensitivity is less than 100%. Celiac disease is almost exclusively associated with the presence of HLA DQ2 and/or HLA DQ8. These HLA genes are not diagnostic of celiac disease, but provide a testing alternative in antibody-negative individuals who meet the clinical diagnostic criteria for celiac disease.

18. A specimen appears to have a perinuclear staining pattern in an antineutrophil cytoplasmic antibody (ANCA) immunofluorescent assay using ethanol-fixed neutrophils, suggesting the possibility of a perinuclear ANCA (pANCA). On which of the following substrates would this specimen display cytoplasmic speckling?

- A.** Formalin-fixed neutrophils
- B.** Unfixed neutrophils
- C.** HEp-2 cells
- D.** Rabbit kidney tissue

Immunology/Select routine laboratory procedures/Autoimmune/ANCA/Testing/2

Answer to Question 18

- 18. A** Antibodies to neutrophil cytoplasmic antigen demonstrating a perinuclear pattern of fluorescence indicate a diagnosis of vasculitis. However, atypical ANCAs and ANAs also demonstrate a perinuclear staining pattern on ethanol-fixed neutrophils. To differentiate these from pANCA, specimens appearing as pANCAs on ethanol-fixed cells are tested on formalin-fixed neutrophils. The myeloperoxidase-containing granules that coalesce around the nuclear membrane during ethanol fixation will remain in the cytoplasm during formalin fixation. Thus, pANCAs will have a cytoplasmic (cANCA) pattern on a formalin-fixed slide, but ANAs will retain a

perinuclear pattern and the fluorescence will be diminished.

3.5

Hypersensitivity

1. Which of the following is a description of a type I hypersensitivity reaction?

- A. Ragweed antigen cross-links with IgE on the surface of mast cells, causing release of preformed mediators and resulting in symptoms of an allergic reaction
- B. Anti-Fy^a from a pregnant woman crosses the placenta and attaches to the Fy^a antigen-positive RBCs of the fetus, destroying the RBCs
- C. Immune complex deposition occurs on the glomerular basement membrane of the kidney, leading to renal failure
- D. Exposure to poison ivy causes sensitized T cells to release lymphokines that cause a localized inflammatory reaction

Immunology/Apply knowledge of fundamental biological characteristics/Hypersensitivity/2

2. Why is skin testing the most widely used method to test for a type I hypersensitivity reaction?

- A. It causes less trauma and is more cost effective than other methods
- B. It has greater sensitivity than in vitro measurements
- C. It is more likely to be positive for IgE-specific allergens compared with other methods
- D. It may be used to predict the development of further allergen sensitivity

Immunology/Apply principles of basic laboratory procedures/Hypersensitivity/Testing/1

3. Which in vitro test measures IgE levels against a specific allergen?

- A. Histamine release assay
- B. Radioimmunosorbent test (RIST)
- C. Fluorescent allergosorbent test (FAST)
- D. Precipitin radioimmunosorbent test (PRIST)

Immunology/Apply principles of basic laboratory procedures/Hypersensitivity/IgE testing/1

4. A patient who is blood group O is accidentally transfused with group A blood and develops a reaction during the transfusion. What antibody is involved in this type II reaction?

- A. IgM
- B. IgE
- C. IgG and IgE
- D. IgG

Immunology/Apply principles of basic laboratory procedures/Hypersensitivity/Testing/1

Answers to Questions 1–4

1. **A** Type I immediate hypersensitivity (anaphylactic) responses are characterized by IgE molecules binding to mast cells via the Fc receptor. Cross-linking of surface IgE caused by binding of allergens causes the mast cell to degranulate, releasing histamine and other chemical mediators of allergy. Answer B describes a type II reaction; C

describes a type III reaction; and D describes a type IV reaction.

2. **B** Skin testing is considered much more sensitive than in vitro tests that measure either total or antigen-specific IgE.
3. **C** FAST is a fluorescent assay that measures specific IgE; RIST and PRIST tests are radioimmunoassays that measure total IgE. The FAST procedure and a chemiluminescent assay have replaced the radioallergosorbent test (RAST), for measuring allergen specific IgE. The histamine release assay is a more general assay.
4. **A** IgG and IgM are the antibodies involved in a type II cytotoxic reaction. Naturally occurring anti-A in the form of IgM is present in the blood of a group O individual and would cause an immediate transfusion reaction. Cell destruction occurs when antibodies bind to cells causing destruction via complement activation, thereby triggering intravascular hemolysis.

5. Which test would measure the coating of RBCs by antibody as occurs in hemolytic transfusion reactions?

- A. Indirect antiglobulin test (IAT)
- B. Direct antiglobulin test (DAT)
- C. ELISA
- D. Hemagglutination

Immunology/Apply principles of basic laboratory procedures/Hemolytic reaction/1

6. Which test detects antibodies that have attached to tissues, resulting in a type-II cytotoxic reaction?

- A. Indirect immunofluorescence
- B. Direct immunofluorescence (DIF)
- C. Immunofixation electrophoresis (IFE)
- D. Hemagglutination

Immunology/Apply principles of basic laboratory procedures/Hemolytic reactions/1

7. Which of the following conditions will most likely result in a false-negative DAT test?

- A. Insufficient washing of RBCs
- B. Use of heavy chain-specific polyclonal anti-human Ig
- C. Use of excessive centrifugal force
- D. Use of a sample obtained by finger puncture

Immunology/Apply knowledge to identify sources of error/Hemolytic reactions/3

8. Which of the following tests will detect circulating immune complexes in the serum of some patients with systemic autoimmune diseases, such as RA?

- A. Direct IFA
- B. Enzyme immunoassay (EIA)
- C. Assay of cryoglobulins
- D. IAT

Immunology/Apply knowledge of fundamental biological characteristics/Hypersensitivity/1

9. What immune elements are involved in a positive skin test for tuberculosis?

- A. IgE antibodies
- B. T cells and macrophages
- C. NK cells and IgG antibody

D. B cells and IgM antibody

Immunology/Apply knowledge of fundamental biological characteristics/Hypersensitivity/1

Answers to Questions 5–9

5. **B** The DAT measures antibody that has already coated RBCs in vivo. DAT and direct IFAs use anti-Ig to detect antibody-sensitized cells.
6. **B** The direct IFA detects the presence of antibody that may cause a type II cytotoxic reaction. For example, renal biopsies from patients with Goodpasture syndrome exhibit a smooth pattern of fluorescence along the basement membrane after reaction with fluorescein isothiocyanate (FITC)–conjugated anti-Ig. The reaction detects antibodies against the basement membrane of the glomeruli.
7. **A** Insufficient washing can cause incomplete removal of excess or unbound Igs and other proteins, which may neutralize the antiglobulin reagent.
8. **C** Most autoimmune diseases involve the formation of antigen–antibody complexes that deposit in tissues, causing local inflammation and necrosis induced by complement activation, phagocytosis, white blood cell (WBC) infiltration, and lysosomal damage. Some patients make monoclonal or polyclonal antibodies with RF activity that bind to serum Igs, forming aggregates that are insoluble at 4°C. These circulating immune complexes are detected by allowing a blood sample to clot at 37°C, transferring the serum to a sedimentation rate tube, and then incubating the serum at 4°C for 3 days.
9. **B** T cells and macrophages are the immune elements primarily responsible for the clinical manifestations of a positive tuberculosis test. Reactions usually take 48 to 72 hours to reach peak development and are characteristic of localized type IV cell-mediated hypersensitivity. The skin reaction is characterized by a lesion containing a mononuclear cell infiltrate.

10. A patient receives a transfusion of packed RBCs and fresh frozen plasma (FFP) and develops an anaphylactic, nonhemolytic reaction. She reports receiving a transfusion 20 years earlier. She had no reaction to the previous transfusion, but she did feel “poorly” a few weeks later. Which of the following transfused substances most likely elicited the reaction?

- A. IgA
- B. Group A antigen
- C. Rho (D) antigen
- D. An antigen belonging to the Duffy system

Immunology/Apply knowledge of fundamental biological characteristics/Immune deficiency/Hypersensitivity/3

11. A patient deficient in the C3 complement component would be expected to mount a normal:

- A. Type I and IV hypersensitivity response
- B. Type II and IV hypersensitivity response
- C. Type I and III hypersensitivity response
- D. Type II and III hypersensitivity response

Answers to Questions 10–11

- 10. A** The fact that this is a nonhemolytic reaction suggests that a non-RBC antigen may be involved. Selective IgA deficiency occurs in approximately one in 700 individuals and is often asymptomatic. Individuals deficient in IgA may make an antibody against the α -heavy chain if they are exposed to IgA via a transfusion. This antibody may lead to a serum sickness reaction if the IgA is still present after antibody formation. This could explain the “poor feeling” the patient had after the initial transfusion. A subsequent transfusion may lead to an Arthus reaction if IgG anti-IgA is present or an anaphylactic reaction if IgE anti-IgA is present.
- 11. A** Complement is involved in types II and III hypersensitivity; thus an individual deficient in C3 will be deficient in those responses. The complement deficiency should have no effect on IgE (type I) or cell-mediated (type IV) hypersensitivities.

3.6

Immunoglobulins, Complement, and Cellular Testing

1. Which of the following symptoms in a young child may indicate an immunodeficiency syndrome?

- A. Anaphylactic reactions
- B. Severe rashes and myalgia
- C. Recurrent bacterial, fungal, and viral infections
- D. Weight loss, rapid heartbeat, breathlessness

Immunology/Apply knowledge of fundamental biological characteristics/Immunodeficiency/1

2. What screening test should be performed first in a young patient suspected of having an immune dysfunction disorder?

- A. Complete blood count (CBC) and WBC differential
- B. Chemotaxis assay
- C. Complement levels
- D. Bone marrow biopsy

Immunology/Apply knowledge of fundamental biological characteristics/Select tests/3

3. Which test should be performed when a patient has a reaction to transfused plasma products?

- A. Ig levels
- B. T-cell count
- C. Hemoglobin levels
- D. RBC enzymes

Immunology/Evaluate laboratory and clinical data to specify additional tests/Select tests/3

4. What is the “M” component in monoclonal gammopathies?

- A. IgM produced in excess
- B. Heavy chain produced in excess
- C. Malignant proliferation of B cells
- D. Monoclonal antibody or cell line

Immunology/Apply knowledge of fundamental biological characteristics/Immunoglobulins/Testing/1

Answers to Questions 1–4

- 1. C** An immunodeficiency syndrome should be considered in a young child who has a history of recurrent bacterial, fungal, and viral infections manifested after the disappearance of maternal IgG. Immunodeficiency disorders may involve deficiencies in production and/or function of lymphocytes and phagocytic cells or a deficiency in production of a complement factor. Choice of laboratory tests is based on the patient's clinical presentation, age, and history.
- 2. A** The first screening tests performed in the initial evaluation of a young patient who is

suspected of having an immune dysfunction are the CBC and differential. WBCs that are decreased in number or abnormal in appearance may indicate further testing.

3. **A** A reaction to plasma products may be found in an IgA-deficient person who has formed anti-IgA antibodies. Ig levels would aid in this determination. Selective IgA deficiency is the most common immunodeficiency disease and is characterized by serum IgA levels below 5 mg/dL. IgA is usually absent from secretions, but the B-cell count is usually normal.
4. **D** The “M” component refers to any monoclonal protein or cell line produced in a monoclonal gammopathy, such as multiple myeloma.

5. **A child suspected of having an inherited humoral immunodeficiency disease is given diphtheria/tetanus vaccine. Two weeks after the immunization, his level of antibody to the specific antigens is measured. Which result is expected for this patient if he, indeed, has a humoral deficiency?**

A. Increased levels of specific antibody
B. No change in the level of specific antibody
C. An increase in IgG-specific antibody but not IgM-specific antibody
D. Increased levels of nonspecific antibody

Immunology/Evaluate laboratory data/Immunoglobulins/Testing/2

6. **Which disease may be expected to show an IgM spike on an electrophoretic pattern?**

A. Hypogammaglobulinemia
B. Multicystic kidney disease
C. Waldenström macroglobulinemia
D. Wiskott-Aldrich syndrome

Immunology/Evaluate laboratory data to make identifications/Immunoglobulins/Testing/2

7. **In testing for DiGeorge syndrome, what type of laboratory analysis would be most helpful in determining the number of mature T cells?**

A. CBC
B. Dihydrorhodamine reduction (DHR) test
C. T-cell mitogen assays
D. Flow cytometry

Immunology/Evaluate laboratory data to make identifications/T cells/Testing/2

8. **Interpret the following description of an IFE assay of urine. Dense wide bands in both the κ - and λ -lanes. No bands present in the heavy-chain lanes.**

A. Normal
B. Light-chain disease
C. Increased polyclonal Fab fragments
D. Multiple myeloma

Immunology/Evaluate laboratory data to make identifications/Immunoglobulins/Testing/2

9. **Free monoclonal light chains are often present in the serum of patients with multiple myeloma and may be useful for disease monitoring. Which of the following assays would be recommended to detect the presence of free light chains in serum?**

A. SPE
B. Urine immunofixation
C. Nephelometry

D. ELISA

Immunology/Evaluate laboratory and clinical data to specify additional tests/Testing/2

Answers to Questions 5–9

5. **B** In a patient with immunodeficiency, the expected levels of specific antibody to the antigens in the vaccine would be decreased or not present. This response provides evidence of deficient antibody production.
6. **C** Waldenström macroglobulinemia is a malignancy of plasmacytoid lymphocytes involving both bone marrow and lymph nodes. The malignant cells secrete monoclonal IgM and are in transition from B cells to plasma cells. In contrast to multiple myeloma, osteolytic bone lesions are not found.
7. **D** DiGeorge syndrome is caused by a developmental failure or hypoplasia of the thymus and results in deficiency of T lymphocytes and cell-mediated immune function. The T-cell count is low, but the level of Igs is usually normal. Flow cytometry is most helpful in determining numbers and subpopulations of T cells.
8. **C** Heavy wide bands seen with both anti- κ and anti- λ antisera indicate excessive polyclonal light-chain excretion. Light-chain disease would show a heavy restricted band for one of the light-chain reactions, but not both. The finding of excess λ - and κ -chains indicates a polyclonal gammopathy with increased Ig turnover and excretion of the light chains as Fab fragments.
9. **C** Free light chains in serum are a sensitive indicator of a monoclonal gammopathy. They are often not present in sufficient quantity to show a band on a protein electrophoresis gel. Detecting light chains in urine is not an indicator of what the serum levels may be. Serum Ig heavy and light chains are most commonly measured by using rate or endpoint nephelometry. ELISA assays are most often used to measure specific antibody levels, not to quantitate Ig heavy- or light-chain isotypes.

10. What is measured in the CH_{50} assay?

- A. RBC quantity needed to agglutinate 50% of antibody
- B. Complement needed to lyse 50% of RBCs
- C. Complement needed to lyse 50% of antibody-sensitized RBCs
- D. Antibody and complement needed to sensitize 50% of RBCs

Immunology/Apply principles of basic laboratory procedures/Complement/Testing/1

11. What type of disorders would show a decrease in C3, C4, and CH_{50} ?

- A. Autoimmune disorders, such as SLE and RA
- B. Immunodeficiency disorders, such as common variable immunodeficiency
- C. Tumors
- D. Bacterial, viral, fungal, or parasitic infections

Immunology/Evaluate laboratory data to make identifications/Complement/Testing/2

12. All of the following tests measure phagocyte function except:

- A. Leukocyte adhesion molecule analysis
- B. DHR assay
- C. Nitro blue tetrazolium (NBT) test

D. IL-2 assay

Immunology/Apply principles of basic laboratory procedures/Phagocytes/Testing/1

Answers to Questions 10–12

- 10. C** CH_{50} is the amount of complement needed to lyse 50% of standardized antibody-sensitized sheep RBCs. It is expressed as the reciprocal of the serum dilution resulting in 50% hemolysis. Low levels are associated with deficiency of some complement components and active systemic autoimmune diseases in which complement is being consumed.
- 11. A** The pattern of decreased C3, C4, and CH_{50} indicates classic pathway activation. This results in consumption of complement and is associated with SLE, serum sickness, subacute bacterial endocarditis, and other immune complex diseases. The inflammatory response seen in malignancy and acute infections gives rise to an increase in complement components. Immunodeficiency caused by an inherited deficiency in complement constitutes only about 1% of immunodeficiency diseases. Such disorders reduce the CH_{50} but involve a deficient serum level of only one complement factor.
- 12. D** The DHR assay and the older NBT tests are used to diagnose chronic granulomatous disease, an inherited disorder in which phagocytic cells fail to kill microorganisms because of a defect in peroxide production (respiratory burst). Leukocyte adhesion deficiency is associated with a defect in the production of integrin molecules on the surface of WBCs and their granules. IL-2 is a cytokine produced by activated T-helper and B cells. It causes B-cell proliferation and increased production of antibody, interferon, and other cytokines. IL-2 can be measured by EIA and is used to detect transplant rejection, which is associated with an increase in the serum and urine levels.

3.7

Tumor Testing and Transplantation

- 1. A patient had surgery for colorectal cancer, after which he received chemotherapy for 6 months. The test for carcinoembryonic antigen (CEA) was normal at this time. One year later, the bimonthly level of CEA was elevated (above 10 ng/mL). An examination and biopsy revealed recurrence of a small tumor. What was the value of the results provided by the CEA test in this clinical situation?**
 - A. Diagnostic information
 - B. Information for further treatment
 - C. Information on the immunologic response of the patient
 - D. No useful clinical information in this case

Immunology/Apply principles of basic laboratory procedures/Tumors/Testing/1

- 2. A carbohydrate antigen 125 assay (CA-125) was performed on a woman with ovarian cancer. After treatment, the levels fell significantly. An examination performed later revealed recurrence of the tumor, but the CA-125 levels remained low. How can this finding be explained?**
 - A. Test error
 - B. CA-125 was the wrong laboratory test; α -fetoprotein (AFP) is a better test to monitor ovarian cancer
 - C. CA-125 may not be sensitive enough when used alone to monitor tumor development
 - D. CA-125 is not specific enough to detect only one type of tumor

Immunology/Apply principles of basic laboratory procedures/Tumors/Testing/3

- 3. What is the correct procedure upon receipt of a test request for human chorionic gonadotropin (hCG) on the serum of a 60-year-old man?**
 - A. Return the request; hCG is not performed on men
 - B. Perform a qualitative hCG test to see if hCG is present
 - C. Perform the test; hCG may be increased in testicular tumors
 - D. Perform the test but use different standards and controls

Immunology/Correlate laboratory data with physiological processes/Tumors/HCG/3

Answers to Questions 1–3

- 1. B** CEA is a glycoprotein that is elevated in about 60% of patients with colorectal cancer and one third or more patients with pulmonary, gastric, and pancreatic cancers. CEA may be positive in smokers and in patients with cirrhosis, Crohn disease, and other nonmalignant conditions. Because sensitivity for malignant disease is low, CEA is not recommended for use as a diagnostic test. However, an elevated CEA level after treatment is evidence of tumor recurrence and the need for second-look surgery.
- 2. C** CA-125 is a tumor associated carbohydrate antigen that is elevated in 70% to 80% of patients with ovarian cancer and about 20% of patients with pancreatic cancer.

Although an increase in CA-125 may indicate recurrent or progressive disease, low levels do not necessarily indicate the absence of tumor growth.

3. **C** hCG is normally tested for in pregnancy; it is increased in approximately 60% of patients with testicular tumors and a lower percentage of those with ovarian, gastrointestinal, breast, and pulmonary tumors. Malignant cells secreting hCG may produce only the β -subunit; therefore, qualitative and quantitative tests that detect both intact hCG and free β -subunits provide better sensitivity than either test by itself.

4. **Would an hCG test using a monoclonal antibody against the β -subunit of hCG likely be affected by an increased level of follicle-stimulating hormone (FSH)?**

- A. Yes, the β -subunit of FSH is identical to that of hCG
- B. No, the test would be specific for the β -subunit of hCG
- C. Yes, a cross reaction would occur because of structural similarities
- D. No, the structure of FSH and hCG are not at all similar

Immunology/Evaluate laboratory data to check for sources of error/hCG/Testing/3

5. **Which of the following substances, sometimes used as a tumor marker, is increased two- or threefold in a normal pregnancy?**

- A. Alkaline phosphatase (ALP)
- B. Calcitonin
- C. Adrenocorticotrophic hormone (ACTH)
- D. Neuron-specific enolase

Immunology/Tumor markers/Testing/1

6. **What is an advantage of performing a prostate-specific antigen (PSA) test for prostate cancer?**

- A. PSA is stable in serum and not affected by digital rectal examination
- B. PSA is increased only in prostatic malignancy
- C. A normal serum level rules out malignant prostatic disease
- D. The percentage of free PSA is elevated in persons with malignant disease

Immunology/Correlate laboratory data with physiological processes/Tumors/PSA/1

7. **Which method is the most sensitive for quantitation of AFP?**

- A. Double immunodiffusion
- B. Electrophoresis
- C. Enzyme immunoassay
- D. Particle agglutination

Immunology/Select appropriate method/AFP/1

Answers to Questions 4–7

4. **B** Luteinizing hormone, FSH, and hCG share a common α -subunit but have different β -subunits. A test for hCG using a monoclonal antibody would be specific for hCG, provided that the antibody was directed against an antigenic determinant on the carboxy terminal end of the β -subunit.
5. **A** Isoenzymes of ALP are sometimes used as tumor markers, but they have a low specificity because they are also increased in nonmalignant diseases. These include placental-like (heat-stable) ALP isoenzymes, which are found (infrequently) in some

malignancies, such as cancer of the lung; bone-derived ALP, which is a marker for metastatic bone cancer; and the fast-migrating liver isoenzyme, which is a marker for metastatic liver cancer. ACTH is secreted as an ectopic hormone in some patients with cancer of the lung. Calcitonin is a hormone produced by the medulla of the thyroid and is increased in the serum of patients with medullary thyroid carcinoma. Neuron-specific enolase is an enzyme that is used as a tumor marker primarily for neuroblastoma.

6. **A** PSA is a glycoprotein with protease activity that is specific for the prostate gland. High levels may be caused by prostate malignancy, benign prostatic hypertrophy, or prostatitis, but PSA is not increased by physical examination of the prostate. PSA has a sensitivity of 80% and a specificity of about 75% for prostate cancer. The sensitivity is sufficiently high to warrant its use in screening tests, but its sensitivity for stage A cancer is below 60%. Most of the serum PSA is bound to protease inhibitors, such as α_1 -antitrypsin and α_1 -antichymotrypsin. Patients with borderline PSA levels (4–10 ng/mL) and a low percentage of free PSA are more likely to have cancer of the prostate compared with patients with a normal percentage of free PSA.
7. **C** AFP is a glycoprotein that is produced in about 80% to 90% of patients with hepatoma and in a lower percentage of patients with other tumors, including retinoblastoma, breast cancer, uterine cancer, and pancreatic cancer. The upper reference limit for serum is only 10 ng/mL, which requires a sensitive method of assay, such as EIA. The high analytical sensitivity of chemiluminescent immunoassays permits detection of reduced AFP levels in maternal serum associated with Down syndrome, as well as elevated levels associated with spina bifida.

8. How is HLA typing used in the investigation of genetic diseases?

- A. For prediction of the severity of the disease
- B. For genetic linkage studies
- C. For direct diagnosis of disease
- D. Is not useful in this situation

Immunology/Correlate clinical and laboratory data/HLA typing/1

9. Select the best donor for a man, blood type AB, in need of kidney transplantation.

- A. His brother, type AB, HLA matched for class II antigens
- B. His mother, type B, HLA matched for class I antigens
- C. His cousin, type O, HLA matched for major class II antigens
- D. Deceased donor, type O, HLA matched for some class I and II antigens

Immunology/Correlate data with other laboratory data to assess test results/Transplantation/Testing/3

10. Interpret the following microcytotoxicity target cell results:

A9 and B12 cells = damaged; A1 and Aw19 cells = intact.

- A. Positive for A1 and Aw19; negative for A9 and B12
- B. Negative for A1 and Aw19; positive for A9 and B12
- C. Error in test system; retest
- D. Impossible to determine

Immunology/Evaluate laboratory data to make identifications/Transplantation/Testing/2

11. Which method, classically used for HLA-D typing, is often used to determine the compatibility between a living organ donor and a recipient?

- A. Flow cytometry
- B. Mixed lymphocyte culture (MLC)
- C. Primed lymphocyte test
- D. Restriction fragment length polymorphism (RFLP)

Immunology/Apply principles of special procedures/Transplantation/HLA typing/1

12. SITUATION: Cells type negative for all HLA antigens in a complement-dependent cytotoxicity assay. What is the most likely cause?

- A. Too much supravital dye was added
- B. Rabbit complement is inactivated
- C. All leukocytes are dead
- D. Antisera is too concentrated

Immunology/Evaluate laboratory data to check for sources of error/HLA typing/3

Answers to Questions 8–12

- 8. B** HLA typing is useful in predicting some genetic diseases and for genetic counseling because certain HLA types show strong linkage to some diseases. HLA typing is not specifically used to diagnose a disease or assess its severity. In linkage studies, a disease gene can be predicted because it is located next to the locus of a normal gene with which it segregates. For example, the relative risk of developing ankylosing spondylitis is 87% in persons who are positive for HLA-B27. Analysis of family pedigrees for the linkage marker and disease can be used to determine the probability that a family member will inherit the disease gene.
- 9. A** A twin or sibling donor of the same blood type and HLA matched for class II antigens is the best donor in this situation. Class II antigens (HLA-D, -DR, -DQ, and -DP) determine the ability of the transplant recipient to recognize the graft. The HLA genes are located close together on chromosome 6, and crossover between HLA genes is rare. Siblings with closely matched class II antigens most likely inherited the same class I genes. The probability of siblings inheriting the same HLA haplotypes from both parents is 1:4.
- 10. B** The microcytotoxicity test is based on the reaction of specific antisera and HLA antigens on test cells. Cells damaged by the binding of antibody and complement are detected with a supravital dye, such as eosin.
- 11. B** Flow cytometry can be used in transplantation to type serologically defined HLA antigens. The one-way mixed lymphocyte reaction is used to identify HLA-D antigens on the donor's lymphocytes and is used for crossmatching living donors with transplant recipients. The assay is time consuming and would not be used as part of a workup for a transplant from a deceased donor. HLA-D incompatibility is associated with the recognition phase of allograft rejection. The primed lymphocyte test is used to identify HLA-DP antigens.
- 12. B** Inactive rabbit complement may not become fixed to antibodies that have bound test leukocytes; therefore, no lysis of cells will occur. When the supravital dye is added, all

cells will appear negative (exclude the dye) for all HLAs.

13. What method may be used for tissue typing instead of serological HLA typing?

- A.** PCR
- B.** Southern blotting
- C.** RFLP
- D.** All of these options

Immunology/Apply principles of special procedures/Transplantation/HLA typing/1

Answer to Question 13

- 13. D** PCR, Southern blotting, and testing for RFLPs may all be used to identify HLA genes. Many laboratories use PCR technology for the routine determination of HLA type.

Immunology Problem-Solving

1. Which of the following serial dilutions contains an incorrect factor?

- A. 1:4, 1:8, 1:16
- B. 1:1, 1:2, 1:4
- C. 1:5, 1:15, 1:45
- D. 1:2, 1:6, 1:12

Immunology/Apply knowledge to recognize sources of error/Serological dilutions/3

2. A patient was tested for syphilis by the RPR method and was reactive. A TP-PA test was performed and the result was negative. Subsequent testing showed the patient to have a high titer of ACAs by the ELISA method. Which routine laboratory test is most likely to be abnormal for this patient?

- A. Activated partial thromboplastin time (APTT)
- B. Anti-smooth muscle antibodies
- C. Aspartate aminotransferase (AST)
- D. C3 assay by immunonephelometry

Immunology/Apply knowledge to recognize sources of error/Anticardiolipin/3

3. Inflammation involves a variety of biochemical and cellular mediators. Which of the following may be increased within 72 hours after an initial infection?

- A. Neutrophils, macrophages, antibody, complement, α_1 -antitrypsin
- B. Macrophages, T cells, antibody, haptoglobin, fibrinogen
- C. Neutrophils, macrophages, complement, fibrinogen, C-reactive protein
- D. Macrophages, T cells, B cells, ceruloplasmin, complement

Immunology/Apply principles of basic immunologic responses/Inflammation/2

Answers to Questions 1–3

- 1. D** All the dilutions are multiplied by the same factor in a progression except the last one: 1:2 to 1:6 is $\times 3$, whereas 1:6 to 1:12 is $\times 2$. Threefold dilutions of a 1:2 dilution would result in a 1:6 followed by a 1:18.
- 2. A** Approximately 50% to 70% of patients with ACAs also have the lupus anticoagulant (LAC) in their serum. LAC is an Ig that interferes with in vitro coagulation tests: prothrombin time (PT), APTT, and dilute Russell viper venom (DRVV) time. These tests require phospholipid for the activation of factor X. About 30% of patients with antibodies to cardiolipin or phospholipids have a biological false-positive RPR result. Anti-smooth muscle is most commonly associated with chronic active hepatitis, and increased AST with necrotic liver diseases. Although ACA and LAC may be associated with SLE, the majority of patients with these antibodies do not have SLE and would have a normal C3 level.

3. **C** The correct list, in which *all* mediators are involved in an inflammatory response within 72 hours after initial infection, is neutrophils, macrophages, complement, fibrinogen, and C-reactive protein. Phagocytic cells, acute phase reactants, and fibrinolytic factors enter the site of inflammation. Antibody and lymphocytes do not enter until later.
4. **An 18-month-old boy has recurrent sinopulmonary infections and septicemia. Bruton thymidine kinase deficiency is suspected. Which test result would be markedly decreased?**
A. Serum IgG, IgA, and IgM
B. Total T-cell count
C. Both B- and T-cell counts
D. Lymphocyte proliferation with phytohemagglutinin stimulation
Immunology/Correlate laboratory data with physiological processes/Immunodeficiency/Testing/2
5. **A patient received 5 units of FFP and developed a severe anaphylactic reaction. He has a history of respiratory and gastrointestinal infections. Post-transfusion studies showed all 5 units to be ABO compatible. What immunologic test would help to determine the cause of this transfusion reaction?**
A. Complement levels, particularly C3 and C4
B. Flow cytometry for T-cell counts
C. Measurement of Igs
D. NBT test for phagocytic function
Immunology/Determine laboratory tests/Immunodeficiency/Testing/3
6. **IFE revealed excessive amounts of polyclonal IgM and low concentrations of IgG and IgA. What is the most likely explanation of these findings and the best course of action?**
A. Proper amounts of antisera were not added; repeat both tests
B. Test specimen was not added properly; repeat both procedures
C. Patient has common variable immunodeficiency; perform B-cell count
D. Patient has immunodeficiency with hyper-M; perform CD40 ligand (CD154) analysis
Immunology/Correlate laboratory data with physiological processes/Immunodeficiency/Testing/3
7. **SITUATION: A 54-year-old man was admitted to the hospital after having a seizure. Many laboratory tests were performed, including an RPR, but none of the results was positive. The physician suspects a case of late (tertiary) syphilis. Which test should be performed next?**
A. Repeat RPR, followed by VDRL
B. Treponemal test, such as TP-PA on serum
C. VDRL on CSF
D. No laboratory test is positive for late (tertiary) syphilis
Immunology/Correlate laboratory data with physiological processes/Syphilis/Testing/3

Answers to Questions 4–7

4. **A** The patient with Bruton thymidine kinase deficiency presents with clinical symptoms related to recurrent infections, demonstrated in the laboratory by decreased or absent

Igs. Peripheral blood B cells are absent or markedly reduced, but T cells are normal in number and function. Because phytohemagglutinin is a T-cell mitogen, the lymphocyte proliferation test using PHA would be normal for this patient.

- 5. **C** The patient had an anaphylactic reaction to a plasma product. This, combined with the history of respiratory and gastrointestinal infections, suggests a selective IgA deficiency. Measurement of Igs would be helpful in this case. A low serum IgA and normal IgG substantiate the diagnosis of selective IgA deficiency. Such patients frequently produce anti-IgA, which is often responsible for a severe transfusion reaction when ABO-compatible plasma is administered.
- 6. **D** Low plasma concentrations of IgG and IgA and an abundance of IgM is consistent with the CD40 ligand deficiency. Most cases are X-linked and result from a mutation of the gene *TNFSF5*, which encodes a receptor needed for switching Ig production. Patients with common variable immunodeficiency have low serum IgG, IgA, and IgM.
- 7. **B** Serum antibody tests, such as RPR and VDRL, are often negative in cases of late syphilis. However, treponemal tests remain positive in greater than 95% of cases. The VDRL test on CSF is the most specific test for diagnosis of neurosyphilis because treponemal test results remain positive after treatment. It should be used as the confirmatory test when the serum treponemal test result is positive. However, the CSF VDRL is limited in sensitivity and would not be positive if the serum treponemal-specific antibody test was negative.

8. **A patient came to his physician complaining of a rash, severe headaches, stiff neck, and sleep problems. Laboratory tests of significance were an elevated sedimentation rate (ESR) and slightly increased liver enzymes. Further questioning of the patient revealed that he had returned from a hunting trip in upstate New York 4 weeks ago. His physician ordered a serological test for Lyme disease, and the assay was negative. What is the most likely explanation of these results?**

- A. The antibody response is not sufficient to be detected at this stage
- B. The clinical symptoms and laboratory results are not characteristic of Lyme disease
- C. The patient likely has early-stage HBV infection
- D. Laboratory error has caused a false-negative result

Immunology/Correlate laboratory data with physiological processes/Lyme testing/Testing/3

9. **A 19-year-old girl came to her physician complaining of a sore throat and fatigue. Upon physical examination, lymphadenopathy was noted. Reactive lymphocytes were noted on the differential, but a rapid test for antibodies to IM was negative. Liver enzymes were only slightly elevated. What test(s) should be ordered next?**

- A. Hepatitis testing
- B. EBV serological panel
- C. HIV confirmatory testing
- D. Bone marrow biopsy

Immunology/Correlate laboratory data with physiological processes/Testing/3

10. **A patient received 2 units of RBCs following surgery. Two weeks after the surgery, the patient was seen by his physician and was found to have mild jaundice and slightly elevated liver enzymes. Hepatitis testing, however, was negative. What should be done next?**

- A. Nothing until more severe or definitive clinical signs develop
- B. Repeat hepatitis testing immediately
- C. Repeat hepatitis testing in a few weeks
- D. Check blood bank donor records and contact donor(s) of transfused units

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/3

11. A hospital employee received the final dose of the hepatitis B vaccine 3 weeks ago. She wants to donate blood. Which of the following results are expected from the hepatitis screen, and will she be allowed to donate blood?

- A. HBsAg, positive; anti-HBc, negative—she may donate
- B. HBsAg, negative; anti-HBc, positive—she may not donate
- C. HBsAg, positive; anti-HBc, positive—she may not donate
- D. HBsAg, negative; anti-HBc, negative—she may donate

Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/3

Answers to Questions 8–11

- 8. A** The antibody response to *B. burgdorferi* may not develop until several weeks after initial infection. The antibody test should be repeated 2 to 4 weeks later. To confirm a positive EIA, samples that initially test positive or equivocal (indeterminate) should be retested using a second enzyme immunoassay or immunoblot method. Regardless of the laboratory results, if the physician suspects Lyme disease, treatment should begin immediately.
- 9. B** An EBV serological panel would give a more accurate assessment than a rapid slide IM test. The time of appearance of the various antibodies to the viral antigens differs according to the clinical course of the infection.
- 10. C** The level of HBsAg may not have reached detectable levels, and antibodies to HBc and HCV would not have yet developed. Waiting 1 or 2 weeks and repeating the tests may reveal evidence of hepatitis virus infection.
- 11. D** She may donate if she is symptom free. The response to hepatitis B vaccine would include a positive result for anti-HBs, a test not normally a part of routine donor testing. She will be negative for HBsAg and anti-HBc; however, transient antigen positivity (less than 2 weeks) may be seen following vaccination.

12. A pregnant woman came to her physician with a maculopapular rash on her face and neck. Her temperature was 37.7°C. Rubella tests for both IgG and IgM antibody were positive. What positive test(s) would reveal a diagnosis of congenital rubella syndrome in her baby after birth?

- A. Positive rubella tests for both IgG and IgM antibody
- B. Positive rubella test for IgM
- C. Positive rubella test for IgG
- D. No positive test is revealed in congenital rubella syndrome

Immunology/Correlate laboratory data with physiological processes/Rubella/Testing/3

13. SITUATION: A patient with RA has acute pneumonia but a negative result on throat culture. The physician suspects an infection with *Mycoplasma pneumoniae* and requests an IgM-specific antibody test. The test is performed directly on serial dilutions of serum

less than 4 hours old. The result is positive, giving a titer of 1:32. However, the test is repeated 3 weeks later, and the titer remains at 1:32. What test should be performed to determine if the patient is truly infected with *M. pneumoniae*?

- A. IgG anti-*M. pneumoniae*
- B. Cold agglutinins
- C. *M. pneumoniae* PCR or other molecular assay
- D. Respiratory culture

Immunology/Apply knowledge to recognize sources of error/IgM testing/3

14. A patient had a PSA level of 60 ng/mL the day before surgery to remove a localized prostate tumor. One week after surgery, serum PSA was determined to be 8 ng/mL by the same method. What is the most likely cause of these results?

- A. Incomplete removal of the malignancy
- B. Cross reactivity of the antibody with another tumor antigen
- C. Testing too soon after surgery
- D. Hook effect with the PSA assay

Immunology/Apply knowledge to recognize inconsistent results/Tumor markers/3

Answers to Questions 12–14

- 12. B** A finding of IgG is not definitive for congenital rubella syndrome because IgG crosses the placenta from the mother; however, demonstration of IgM, even in a single neonatal sample, is diagnostic.
- 13. C** The IgM-specific antibody test for *M. pneumoniae* detects antibodies to mycoplasmal membrane antigens and, unlike cold agglutinins, is specific for *M. pneumoniae*. A positive result (titer of 1:32 or higher) occurs during the acute phase in about 87% of *M. pneumoniae* infections and does not need to be confirmed by assay of convalescent serum. However, *Mycoplasma* IgM may last a year or more, thus its presence does not always indicate a current infection. PCR performed on a respiratory specimen is the definitive test and should be performed if there is a question about an IgM result.
- 14. C** When monitoring the level of a tumor marker for treatment efficacy or recurrence, the half-life of the protein must be considered when determining the testing interval. PSA has a half-life of almost 4 days and would not reach normal levels after surgery for approximately 3 to 4 weeks. The hook effect is the result of very high antigen levels giving a lower than expected result in a double antibody sandwich assay when both antibodies and sample are added at the same time.

15. A patient with symptoms associated with SLE and scleroderma was evaluated by immunofluorescence microscopy for ANAs by using the HEp-2 cell line as substrate. The cell line displayed a mixed pattern of fluorescence that could not be separated by serial dilutions of the serum. Which procedure would be most helpful in determining the antibody profile of this patient?

- A. Use of a different tissue substrate
- B. Absorption of the serum using the appropriate tissue extract
- C. Requesting a new specimen
- D. ELISA tests for specific antibodies

Immunology/Apply knowledge to identify laboratory tests/ANA/Testing/3

16. A patient with joint swelling and pain tested negative for serum RF by both latex agglutination and ELISA methods. What other test would help establish a diagnosis of RA in this patient?

- A. Anti-CCP
- B. ANA testing
- C. Flow cytometry
- D. Complement levels

Immunology/Correlate laboratory data with physiological processes/RA/Testing/3

17. What is the main advantage of the recovery and reinfusion of autologous stem cells?

- A. It slows the rate of rejection of transplanted cells
- B. It prevents graft-versus-host disease
- C. No HLA testing is required
- D. Engraftment occurs in a more efficient sequence

Immunology/Apply knowledge of fundamental biological characteristics/Transplantation/2

18. A transplant recipient began to show signs of rejection 8 days after transplantation, and the organ was removed. What immune elements might be found in the rejected organ?

- A. Antibody and complement
- B. Primarily antibody
- C. Macrophages
- D. T cells

Immunology/Correlate laboratory data and basic immune response/Transplantation/Rejection/3

Answers to Questions 15–18

15. D Many patients with multiorgan autoimmune disease display symptoms that overlap two or more diseases and have complex mixtures of serum autoantibodies. The HEp-2 substrate is the most sensitive cell line for immunofluorescent microscopy because it contains cells in various mitotic stages, which exposes the serum to more antigens. Use of a nonhuman substrate, such as *Crithidia*, may help identify dsDNA antibodies but would not aid in differentiating all of the antibodies in a complex mixture. The best methods are ELISA, line blots, and multiplex tests because they are more specific than immunofluorescence microscopy for identifying antibodies to specific antigens. These assays are often used to measure antibodies to ENAs, which may be partially or completely lost during fixation of cells used for immunofluorescent microscopy. These antibodies cause a speckled pattern and are seen in a wide range of autoimmune diseases. Identification of the anti-ENA specificities is helpful in differentiating these diseases.

16. A Antibodies to CCP are often found in RF-negative patients with RA. The absence of RFs from serum does not rule out a diagnosis of RA, and more than half the patients who are diagnosed with RA present initially with a negative serum RF result. Both RF and anti-CCP are criteria assays for diagnosing RA, and at least one of them must be positive for a confirmed diagnosis.

17. B The main advantage to the patient from the reinfusion of autologous stem cells is that

the procedure prevents graft-versus-host disease, especially in the immunocompromised patient. Although HLA testing is not required, this is not the primary advantage for patient care.

- 18. D** Acute rejection occurs within 3 weeks of transplantation. The immune element most likely to be involved in an acute rejection is the T cell in a type IV, delayed hypersensitivity (cell-mediated) reaction. Preformed antibody, and possibly complement, is usually involved in hyperacute (immediate) rejection and chronic rejection.

- 19. A** A patient with ovarian cancer who has been treated with chemotherapy is being monitored for recurrence by using serum CA-125, CA-50, and CA 15–3. Six months after treatment the CA 15–3 is elevated, but the CA-125 and CA-50 remain low. What is the most likely explanation of these findings?

- A. Ovarian malignancy has recurred
- B. CA 15–3 is specific for breast cancer and indicates metastatic breast cancer
- C. Testing error occurred in the measurement of CA 15–3 caused by poor analytical specificity
- D. The CA 15–3 elevation is spurious and probably benign

Immunology/Correlate laboratory data with physiological processes/Tumor markers/Testing/3

- 20. An initial and repeat fourth-generation HIV screening test are both positive. The antibody differentiation assay is negative, as is the qualitative RNA PCR test. The patient shows no clinical signs of HIV infection, and the patient's CD4 T-cell count is normal. Based on these results, which conclusion is correct?**

- A. Patient is diagnosed as HIV-1 positive
- B. Patient is diagnosed as HIV-2 positive
- C. Results are inconclusive
- D. Patient is diagnosed as HIV-1 negative

Immunology/Apply knowledge to recognize inconsistent results/HIV/3

- 21. A woman who has been pregnant for 12 weeks is tested for toxoplasmosis. Her IgM ELISA titer is 2.6 (reference range less than 1.6), and her IgG ELISA value is 66 (reference range less than 8). The physician asks you if these results indicated an infection during the past 12 weeks. Which of the following tests would you recommend to determine if the woman was infected during her pregnancy?**

- A. Toxoplasmosis PCR on amniotic fluid
- B. Toxoplasmosis IgM on amniotic fluid
- C. Toxoplasmosis IgG avidity
- D. Amniotic fluid culture

Immunology/Correlate laboratory data with physiological processes/Time course of immune response/Toxoplasmosis/Testing/3

Answers to Questions 19–21

- 19. A** Although CA-125 is the most commonly used tumor marker for ovarian cancer, not all ovarian tumors produce CA-125. Greatest sensitivity in monitoring for recurrence is achieved when several markers known to be increased in the malignant tissue type are

measured simultaneously and when the markers are elevated (by malignancy) prior to treatment. In addition to limited sensitivity, no single tumor marker is entirely specific. Carbohydrate and other oncofetal antigens are produced by several malignant and benign conditions. Although testing errors may occur in any situation, measurements of carbohydrate antigens use purified monoclonal antibodies with very low cross reactivities.

20. **D** The fourth-generation algorithm detects infection approximately 2 weeks after exposure. This patient has negative follow-up test results for both antibody and nucleic acid, so she would be considered HIV negative. The lack of symptoms would not be consistent with a very recent infection; the patient would likely have an acute retroviral syndrome. However, a repeat test may be performed if clinical suspicion remains high.
21. **C** Although IgM is positive, in toxoplasmosis, specific IgM may remain detectable for a year or more following infection. IgG avidity, or the strength of binding of a serum to the antigen of interest, is a useful method to determine if an infection is recent or in the distant past. IgG avidity will increase with time following an infection. Amniotic fluid testing is not useful for determining when the mother might have been infected.
22. **On January 4, an SPE on a specimen obtained at your hospital in North Dakota from a 58-year-old patient shows a band at the β — γ junction. The specimen was also positive for RF. You recommend that an immunofixation test be performed to determine if the band represents a monoclonal Ig. Another specimen is obtained 2 weeks later by the physician in his office 30 miles away, and whole blood is submitted to you for IFE. The courier placed the whole blood specimen in an ice chest for transport. In this specimen, no β — γ band is seen in the serum protein lane, and the IgM lane is very faint. The RF on this specimen was negative. The physician wants to know what went wrong in your laboratory. Your response is:**
- A. Nothing went wrong in our laboratory; the patient had an infection 2 weeks ago, and it had cleared up
 - B. Something went wrong in our laboratory—we likely mislabeled one of the specimens; please resubmit a new specimen, and we will test it at no charge
 - C. We will run a second specimen after 2-mercaptoethanol treatment, which will eliminate IgM aggregates and allow for more sensitive monoclonal IgM detection
 - D. Please redraw another specimen from the patient, and this time, separate the serum from the clot in your office before placing the specimen on ice and sending it to us by courier
- Immunology/Correlate laboratory data with physiological processes/Specimen integrity/3*
23. **A patient undergoing dialysis is positive for both HBsAg and anti-HBs. The physician suspects a laboratory error. Do you agree?**
- A. Yes; the patient should not test positive for both HBsAg and anti-HBs
 - B. No; incomplete dialysis of a patient in the core window phase of HBV infection will yield this result
 - C. No; it is likely the patient has recently received a hepatitis B booster vaccination within the past week, and this could have caused these results
 - D. Perhaps; a new specimen should be submitted to clear up the confusion
- Immunology/Correlate laboratory data with physiological processes/Hepatitis/Testing/3*
24. **You are evaluating an ELISA assay as a replacement for your IFA ANA test. You test**

50 specimens in duplicate on each assay. The ELISA assay uses a HEP-2 extract as its antigen source. The correlation between the ELISA and IFA tests is only 60% (30 of 50 specimens agree). Which of the following is the next best course of action?

- A.** Test another 50 specimens
- B.** Perform a competency check on the medical laboratory scientists who performed the tests
- C.** Order a new lot of both kits and then retest on the new lots
- D.** Refer the discrepant specimens for testing by another method

Immunology/Management principles/Method comparison/3

Answers to Questions 22–24

- 22. D** The most likely cause of the discrepant results is the presence of a type II cryoglobulin. This is a monoclonal RF. The protein likely precipitated during the courier ride and was, thus, in the clot when the laboratory separated the serum.
- 23. C** HBsAg will remain detectable at low levels following a vaccination for up to 1 to 2 weeks. Thus, patients who have received a second injection of hepatitis B vaccine may have anti-HBs and detectable antigen for a brief period. This has been reported more frequently in patients undergoing dialysis and in pediatric populations.
- 24. D** In this situation, you have already tested the specimens in duplicate. Testing an additional 50 specimens will not change the fact that you have 20 discrepant specimens. The best course of action is to determine what antibodies are actually present in these specimens. Then, you can determine whether ELISA or IFA is a better procedure for detecting the most clinically relevant antibodies. You could perform clinical chart reviews as an alternative, but obtaining that data would be difficult and much of it may be subjective.

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